Contents

Chapter 1

GAUSS'S LAW

Coulomb's law can always be used to calculate the electric field $\vec{E}$ for any discrete or continuous distribution of charges at rest. The sums or integrals might sometimes be complicated to calculate the resultant field at any point. In the chapter, we wish to discuss an alternative to Coulomb's law, called Gauss's law that provides a more useful and instructive approach to calculate the electric field in situations having certain symmetries. Although Gauss's law and Coulomb's law give identical results in the cases in which both can be used, Gauss's law is considered a more fundamental equation than Coulomb's law.

1.1 The Electric Field

Every body have an idea about *gravitational field*. Around the earth surface there is a space region with existence of one kind of force field known as *gravitational field*. If a body falls through the earth's gravitational field, it experience one kind of force expressed by the magnitude

$$\vec{F} = m_0\vec{g}, \tag{1.1}$$

$$\text{where,} \quad \vec{g} = \frac{\vec{F}}{m_0}. \tag{1.2}$$

The field $\vec{g}$ is a vector whose direction gives the direction of the gravitational force at that point and whose magnitude indicates the **"strength"** of the gravitational effect at that point.

Similarly around an electric charged particle or particles, there is a space region where one kind of force field exists. *This force field existing around the electric charged particle or particles is called electric field and denoted by* $\vec{E}$. If a test charge q_0 is placed such kind of electric field $\vec{E}$, in analogy with gravitational field Eq. 1.1, it also experienced a kind of force and can be expressed as

$$\vec{F} = q_o\vec{E}, \tag{1.3}$$

which gives the electric field strength can be written as

$$\vec{E} = \frac{\vec{F}}{q_o}. \tag{1.4}$$

The direction of the vector $\vec{E}$ is the same as the direction of $\vec{F}$.

1.2 Electric Field Lines

The concept of the electric field was introduced in the early 19th century by **Michael Faraday**. He

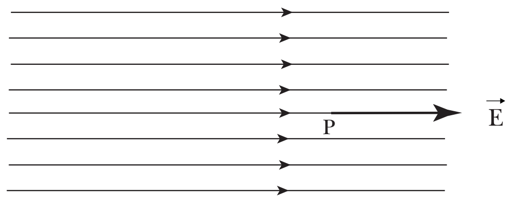


Figure 1.1: Electric field lines for a uniform electric field and direction at any point.

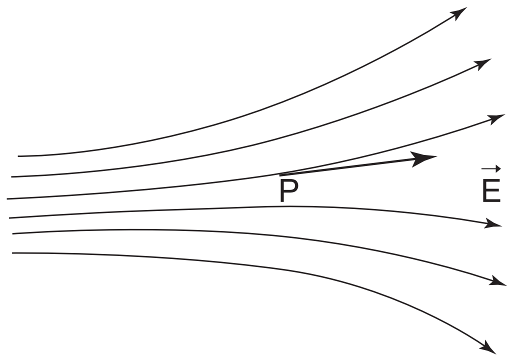


Figure 1.2: Nonuniform electric field lines and direction at any point.

developed a graphical representation, in which he imagined the space around an electric charge to be filled with *lines of force*. According to his concept of force lines we shall try to visualize some pattern of *electric field lines*.

Direction of the field line: *The tangent to the electric field line passing through any point in space gives the direction of the electric field at that point,* as it is shown in Figs. 1.1, 1.2, 1.3 and 1.4.

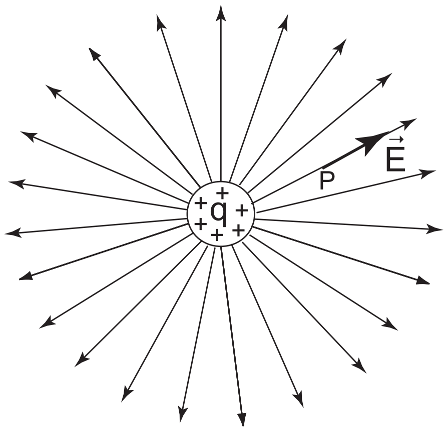


Figure 1.3: Electric field lines surrounding a group of point charges and direction at any point.

Figure 1.1 shows the electric force lines which represents a uniform electric field. Note that the lines are parallel and equally spaced to each other.

Figure 1.2 shows nonuniform electric field lines. At any point P the direction of field $\vec{E}$ has been shown by an arrowhead marker. Figure 1.3 represents the field lines for an isolated group of positive point charges. The electric field lines start on positive charges and point radially outward. If the charge were negative, the field lines would point in the opposite direction i.e. radially inward. Figure 1.4 shows the electric field lines for an electric dipole. The electric field lines start on positive charge and end on negative charge.

Magnitude of the electric field: ***The magnitude of the electric field at any point is proportional to the number of field lines per unit cross sectional area perpendicular to the lines.***

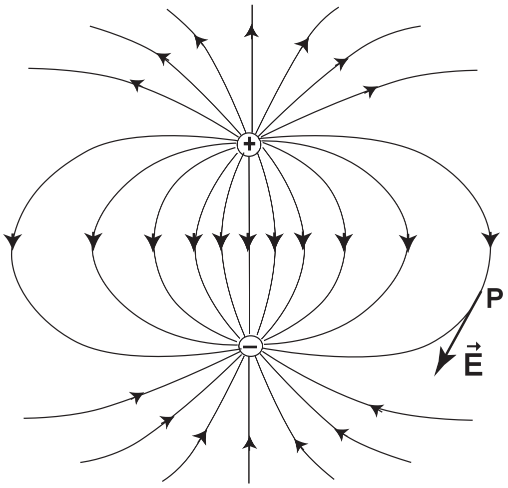


Figure 1.4: Lines of electric field for an electric dipole and direction at any point.

1.3 The Flux of the Electric Field

The 'flux' comes from a Latin word meaning 'to flow.' The symbol of the flux is written by Φ. It is a property of any vector field that refers to a hypothetical surface in the field, which may be closed or open. **The flux of a vector field to be a measure of the flow or penetration of the field vectors through an imaginary surface in the field. For an electric field the flux Φ_E is a measure of the**

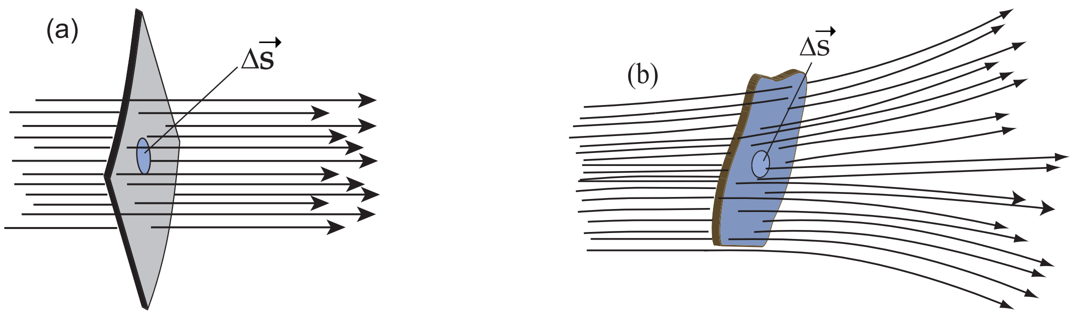


Figure 1.5: (a) Surface in a uniform electric field $\vec{E}$, (b) Surface in a nonuniform electric field $\vec{E}$.

number of lines of the electric field that passes through a surface. Figure 1.5 shows an arbitrary surface immersed in a uniform and nonuniform electric field. Let us divide the surface into small squares of area Δs, each of which is small enough so that it may be considered to be plane. Each element of area can be represented as a vector $\Delta\vec{s}$, whose magnitude to be the area Δs.

The total flux of any electric field over the surface can be calculated by the definition as

$$\Phi_E = \sum \vec{E}.\Delta\vec{s}. \tag{1.5}$$

Replacing the sign of the summation by an integral sign over the surface the Eq. 1.5 yields

$$\Phi_E = \int \vec{E}.d\vec{s}. \tag{1.6}$$

If the angle between the vectors $\vec{E}$ and $d\vec{s}$ be θ then the Eq. 1.6 can be rewritten as

$$\Phi_E = \int E\ ds\ \cos\theta. \tag{1.7}$$

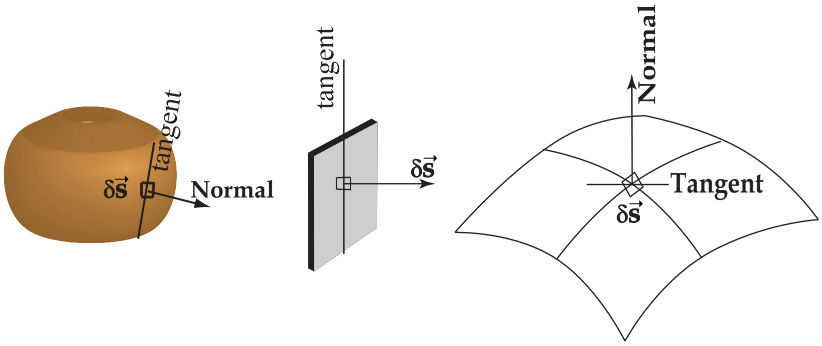


Figure 1.6: Direction of a surface element.

Direction of a surface:

The direction of a small element of surface $\Delta\vec{s}$ is shown in Fig. 1.6. *The direction of a surface element can be directed at any point by a line drawn perpendicularly on the tangent of the surface element.*

Flux through a closed surface in a uniform field:

Let us consider a hypothetical closed cylindrical surface is immersed in a uniform electric field, as it is shown in Fig. 1.7. The axis of the cylinder is parallel to the electric lines of force. Calculate the flux Φ_E for this closed surface.

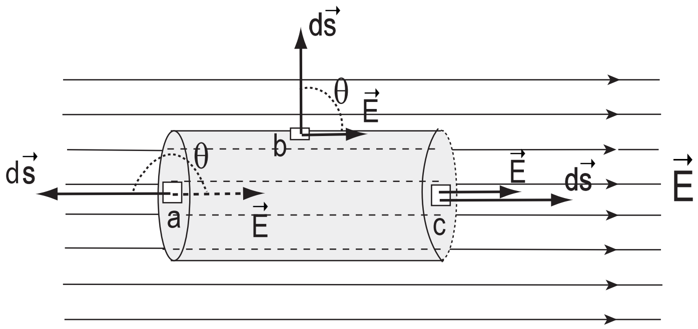


Figure 1.7: A closed cylindrical surface immersed in a uniform electric field.

The flux Φ_E can be calculated by the formula which is applicable for a closed surface,

$$\Phi_E = \oint \vec{E}.d\vec{s}.$$

The integral can be perform by the sum of three parts of integration over the left cylindrical cap 'a,' the curved cylindrical surface 'b' and the right cylindrical cap 'c.'

Therefore,

$$\begin{aligned}
\Phi_E &= \oint \vec{E}.d\vec{s},\\
&= \int_a \vec{E}.d\vec{s} + \int_b \vec{E}.d\vec{s} + \int_c \vec{E}.d\vec{s},\\
&= \int_a E\ ds\ \cos\theta + \int_b E\ ds\ \cos\theta + \int_c E\ ds\ \cos\theta,\\
&= \int_a E\ ds\ \cos 180^o + \int_b E\ ds\ \cos 90^o + \int_c E\ ds\ \cos 0^o,
\end{aligned}$$

$$= -ES_a + 0 + ES_b.$$

Since the surface area S_a is equal to the surface area S_b,

$$\therefore \quad \Phi_E = 0.$$

We find that the net flux enclosed by the cylindrical surface is zero. Because lines of $\vec{E}$ enter at the left and emerge at the right, and there is no electric charge enclosed by the surface.

1.4 Gauss's Law

Consider an imaginary closed surface which contain some electric charge inside it (see Fig.1.8). Gauss's law gives the relation between the flux that cut through a surface and the net charge enclosed by the surface. The imaginary surface is called ***Gaussian surface***. Therefore, Gauss's law which relates the total flux Φ_E through this surface to the net charge q enclosed by the surface, can be written as

$$\Phi_E \propto q,$$

$$\Rightarrow \quad \Phi_E = \frac{1}{\varepsilon_0} q$$

$$\text{or} \qquad \varepsilon_0 \Phi_E = q, \tag{1.8}$$

$$\text{or} \qquad \varepsilon_0 \oint \vec{E}.d\vec{s} = q. \tag{1.9}$$

The circle on the integral sign indicates that the integral is to be carried out over a closed surface.

Here, ε_0 is a constant and its value is $8.854 \times 10^{-12}\ C^2/N - m^2$.

In special case, if there is no charge inside the Gaussian surface, from the Eq. 1.8 we have

$$\Phi_E = 0.$$

The same results has already been found in previous section belongs to the the topic of flux calculation in a uniform field, where also no net charge enclosed that surface. The integral in Eq. 1.9 essentially counts the number of field lines passing through the surface. It is entirely reasonable that the number of field lines passing through a surface should be proportional to the net charge enclosed by the surface.

1.5 Coulomb's Law and Gauss's Law

Coulomb's law can be deduced from Gauss's law. Consider a group of positive charge of amount $+q$. We can apply Gauss's law to calculate electric field E at any point situated at a distance r from the charges. To do so, let us consider an imaginary spherical Gaussian surface with radius r by keeping the charges at the center, as it is shown in Fig. 1.8. The electric field lines $\vec{E}$ should certainly be directed radially outward. The direction of electric field $\vec{E}$ and surface element $d\vec{s}$ must be perpendicular to the spherical surface. Therefore, the angle θ between $\vec{E}$ and $d\vec{s}$ is zero everywhere on the surface.

Using Gauss's law

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = \varepsilon_0 \oint E\ ds\ \cos 0^o = q,$$

$$\text{or} \qquad \varepsilon_0 \oint E\ ds = q.$$

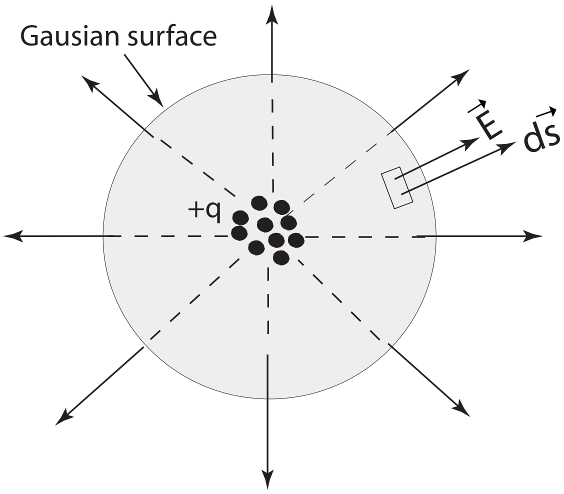


Figure 1.8: A group of positive point charge surrounded by a spherical Gaussian surface.

Since E has the same magnitude for all points on the sphere, it can be termed as constant for integration.

$$\therefore \quad \varepsilon_0 E \oint ds = q,$$

where $\oint ds$ is merely the total surface area of the sphere, $4\pi r^2$. We, therefore, obtain after integration

$$\varepsilon_0 E(4\pi r^2) = q,$$

$$\text{or} \qquad E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}. \tag{1.10}$$

Here, $\frac{1}{4\pi\varepsilon_0}$ is a constant and its value is $8.99 \times 10^9\ N - m^2/C^2$.

Equation 1.10 gives the magnitude of the electric field $\vec{E}$ at any point a distance r from an isolated point charge q and is identical the result of Coulomb's law. Thus by choosing a Gaussian surface with the proper symmetry, one can obtain Coulomb's law from Gauss's law. The two laws can be regarded as equivalent for our application, *Gauss's law is more generally applicable and is regarded as more fundamental equation of electromagnetism.*

1.6 Applications of Gauss's Law

Gauss's law is suitable to calculate $\vec{E}$ if the symmetry of the charge distribution is high enough. One example of this calculation, the field of a point charge, has already been discussed in section 1.5. Next we will discuss some other examples.

1.6.1 Infinite line of charge

Figure 1.9 shows a part of infinite line of charge of constant linear charge density λ. That is, λ is the charge per unit length. We have to calculate E at a distance r from the line. Since the charge distribution is infinitely long, the electric field always be perpendicular to the charge distribution and having cylindrical symmetry. For this argument we can choose a cylindrical Gaussian surface of radius r and length h.

The field $\vec{E}$ is constant with perpendicular direction of line of charge. Now apply Gauss's law,

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q.$$

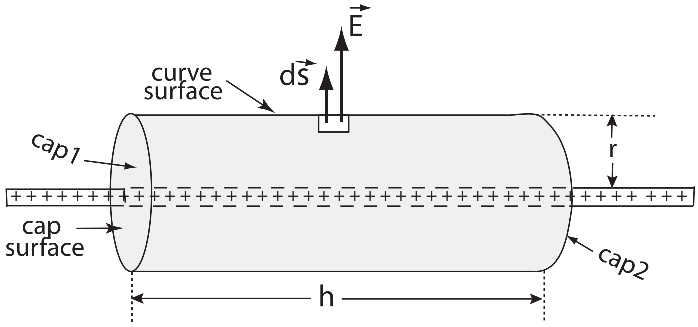


Figure 1.9: Infinite line of charge surrounded by a spherical Gaussian surface.

The left hand side can be subdivided as the integral

$$\varepsilon_0 \left(\int_{curve} E\ ds\ \cos 0^o + \int_{cap1} E\ ds\ \cos 90^o + \int_{cap2} E\ ds\ \cos 90^o \right) = q,$$

$$\text{or} \qquad \varepsilon_0 E(2\pi r h) + 0 + 0 = \lambda h,$$

$$\text{or} \qquad E = \frac{\lambda}{2\pi\varepsilon_0 r}. \tag{1.11}$$

Note that the calculation is much simpler than that of other integration method.

1.6.2 Infinite sheet of charge

Figure 1.10 shows a portion of a thin, nonconducting, infinite sheet of charge of constant positive surface charge density σ (charge per unit area). We have to calculate the electric field at points near the sheet. let us choose a conventional cylindrical surface of cross sectional area S as it is shown in Fig. 1.10.

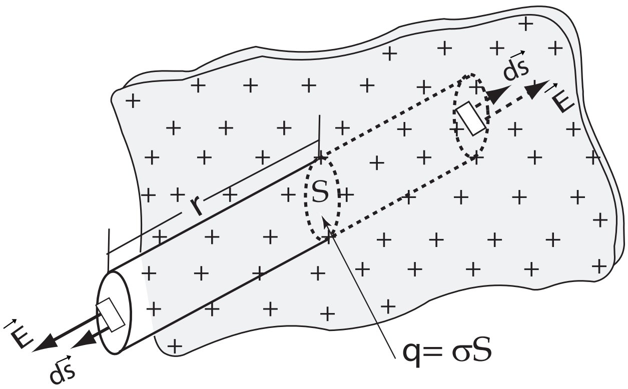


Figure 1.10: An infinite sheet of charge density σ. A conventional cylindrical Gaussian surface is considered here.

Since the sheet is infinitely extended, the electric field should be directed perpendicularly from the plane of the sheet. Only the flux regarding to the two cap surfaces will be effective in the Gaussian surface. Gauss's law gives

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q.$$

Now the surface integral of the L.H.S can be written as

$$\varepsilon_0 \left(\int_{curve} E\ ds\ \cos 90^o + \int_{cap1} E\ ds\ \cos 0^o + \int_{cap2} E\ ds\ \cos 0^o \right) = \sigma S,$$

$$\text{or} \qquad \varepsilon_0(0+ES+ES) = \sigma S,$$
$$\text{or} \qquad 2\varepsilon_0 ES = \sigma S,$$
$$\text{or} \qquad E = \frac{\sigma}{2\varepsilon_0}. \tag{1.12}$$

Note that E is independent to the distance of the points from the charge sheet. So E is the same for all points on each side of the sheet. Although an infinite sheet of charge cannot exist physically, the derivation of Eq. 1.12 gives approximately correct results for real (not infinite) sheets.

1.6.3 A spherical shell of charge

Figure 1.11 shows a thin spherical shell on which a charge q is uniformly distribution. We calculate E for (i) $r > R$ and (ii) $r < R$.

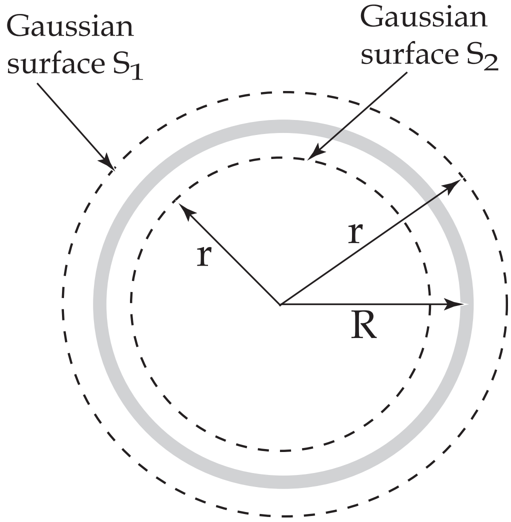


***Figure 1.11:** A spherical shell of charge q with radius R. Gaussian surface with radius $r > R$ and Gaussian surface with radius $r < R$.*

(i) First we calculate E at a distance when $r > R$. In this condition we constitute a Gaussian surface S_1 outside the charge distribution. Applying Gauss's law

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q.$$

For symmetric charge distribution we consider E is also symmetric. For $r > R$ Gauss's law gives

$$\varepsilon_0 E(4\pi r^2) = q$$
$$\text{or} \qquad E_{r>R} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}, \tag{1.13}$$

which gives the same result with Eq. 1.10. Thus the uniformly charged shell behaves like a point or group of point charges for all points outside the shell.

(ii) Now we calculate E at a distance when $r < R$. We constitute a Gaussian surface S_2 inside the charge distribution. Applying Gauss's law

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q.$$

Since there is no charge enclosed by the Gaussian surface, that is $q = 0$ for Gaussian surface S_2. So we have E for inside the shell is as,

$$E_{r<R} = 0. \tag{1.14}$$

The electric field therefore vanishes inside a uniform shell of charge.

1.6.4 Spherical symmetric charge distribution

Figure 1.12 shows a spherical charge distribution with charge density ρ (the charge per unit volume). Radius of the sphere be R. Let us consider the charge density ρ is not constant everywhere of the sphere, but restrict that ρ at any point be radial symmetry. Let us find an expression of E for points (a) outside and (b) inside the charge distribution.

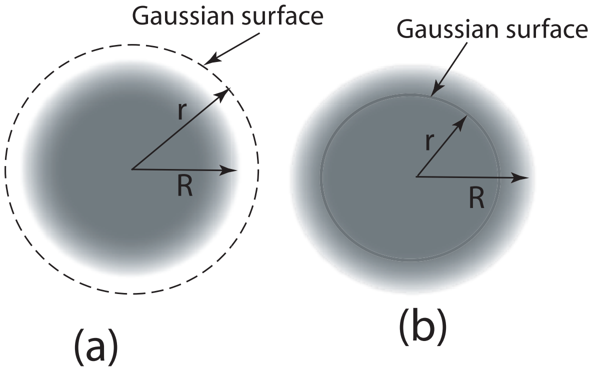


Figure 1.12: A spherical charge distribution of radius R. (i) Gaussian surface with radius $r > R$, (ii) Gaussian surface with radius $r < R$.

(i). Let us consider a spherical Gaussian surface of radius r, where $r > R$ as shown in Fig. 1.12a. Now using Gauss's law we have

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q,$$

$$\text{or} \qquad \varepsilon_0 E(4\pi r^2) = q,$$

$$\text{or} \qquad E_{r>R} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \tag{1.15}$$

We calculate q as

$$q = \rho\left(\frac{4}{3}\pi R^3\right).$$

Then Eq. 1.15 gives the form as

$$E_{r>R} = \frac{\rho R^3}{3\varepsilon_0 r^2}. \tag{1.16}$$

(ii). Let us consider a spherical Gaussian surface of radius r, where $r < R$ as shown in Fig. 1.12b. Now applying Gauss's law we have

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q',$$

$$\text{or} \qquad \varepsilon_0 E(4\pi r^2) = q',$$

$$\text{or} \qquad E_{r<R} = \frac{1}{4\pi\varepsilon_0}\frac{q'}{r^2}. \tag{1.17}$$

We calculate q' as

$$q' = \rho\left(\frac{4}{3}\pi r^3\right).$$

Then Eq. 1.17 gives the form as

$$E_{r<R} = \frac{\rho r}{3\varepsilon_0}. \tag{1.18}$$

We see if $r = 0$, then E vanishes at the center of the sphere. Let us consider a point on the surface of the sphere. That is $r = R$, then Eqs. 1.16 and 1.18 coincide the same result.

1.7 Gauss Law for Dielectric Materials

In this section, it is discussed how an electric field modified in a capacitor and calculated Gauss,s law for a dielectric material. This modification is occurred due to polarization which is not the same as in vacuum.

Let consider an external electrical field E_0 is applied in a parallel plate capacitor without dielectric [Fig. 1.13(a)]. Positive charges $+q$ and negative charges $-q$ are accumulated in the two different plates as it is shown in Fig. 1.13 Consider the Gaussian surface of positive charges.

The Gauss's law in vacuum:

$$\text{Differntial form} \qquad \vec{\nabla}.\vec{E}_0 = \frac{\rho}{\varepsilon_0}. \tag{1.19}$$

$$\text{Integration form} \qquad \oint \vec{E}_0.d\vec{S} = \frac{q}{\varepsilon_0}. \tag{1.20}$$

From Eq. 1.20

$$E_0 A = \frac{q}{\varepsilon_0},$$

The electric field in vacuum

$$E_0 = \frac{q}{\varepsilon_0 A}. \tag{1.21}$$

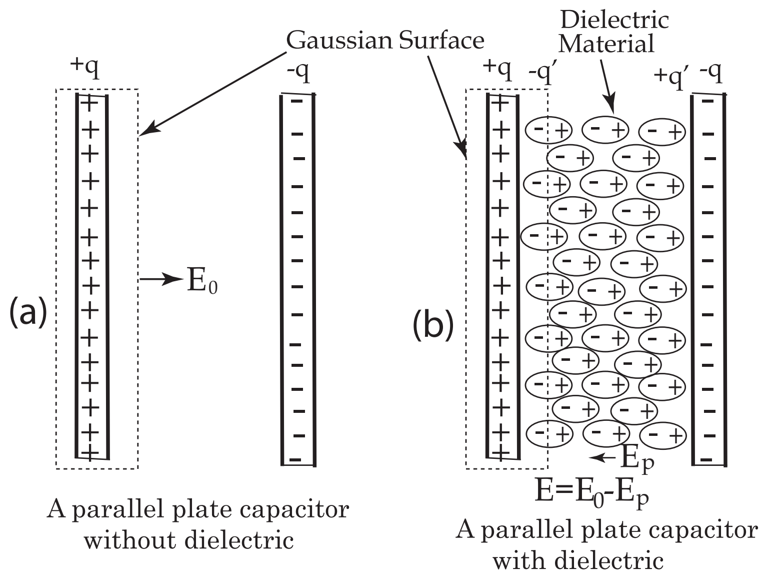


Figure 1.13: A parallel plate capacitor (a) without dielectric, (b) with dielectric.

Consider a dielectric material of dielectric constant κ and permittivity[1] ε is inserted between the plates [Fig. 1.13(b)]. There is a polarization of charges occurred in the dielectric medium. As a result induced charges $-q'$ and $+q'$ are accumulated near the surfaces of the plates.

In the present of dielectric, the Gaussian surface encloses two types of charge: Free charge $+q$ and induced charge $-q'$. Let E be the resultant field within the dielectric. According to Gausss law

$$\begin{aligned} \oint \vec{E}.d\vec{S} &= \frac{q-q'}{\varepsilon_0}, \qquad (1.22) \\ EA &= \frac{q-q'}{\varepsilon_0}, \end{aligned}$$

[1]Permittivity is the ability to resist external electric field of a dielectric. This means a substance with high permittivity requires high external electric field to polarize.

$$E = \frac{q}{\varepsilon_0 A} - \frac{q'}{\varepsilon_0 A}. \tag{1.23}$$

The induced charges ($-q'$ and $+q'$ near both the plates) produce their own field which opposes the external field E_0. It is clear that $E_0 > E$. Now, $E_0/E = \kappa$, where κ is dielectric constant. Therefore, from Eq. 1.21

$$\kappa = \frac{q}{\varepsilon_0 AE} \quad \text{or} \quad E = \frac{q}{\varepsilon_0 A\kappa}.$$

Now from Eq. 1.23

$$\begin{aligned} \frac{q}{\varepsilon_0 A\kappa} &= \frac{q}{\varepsilon_0 A} - \frac{q'}{\varepsilon_0 A} \\ \frac{q}{\kappa} &= q - q' \\ q' &= q\left(1 - \frac{1}{\kappa}\right). \end{aligned} \tag{1.24}$$

Equation 1.24 shows that the induced charge q' is less than the free charge q. Substituting the value $q - q' = \frac{q}{\kappa}$ in Eq. 1.22 one get

$$\oint \vec{E}.d\vec{S} = \frac{q}{\varepsilon_0\kappa} \tag{1.25}$$

Now we known, the dielectric constant is the ratio of permittivity of medium to that of the vacuum i.e, $\kappa = \varepsilon/\varepsilon_0$. From Eq. 1.25

$$\begin{aligned} \oint \vec{E}.d\vec{S} &= \frac{q}{\varepsilon} \\ \text{or} \quad \oint \varepsilon\vec{E}.d\vec{S} &= q \\ \text{or} \quad \oint \vec{D}.d\vec{S} &= q \end{aligned} \tag{1.26}$$

where $\varepsilon E = D$ and is called the displacement vector. Equation 1.26 is the Gauss's law in the presence of dielectric.

1.7.1 Three electric vectors

Electric displacement denoted by D is the charge per unit area that would be displaced across a layer of conductor placed across an electric field. It is also known as electric flux density. Here total charge density $\rho = \rho_f + \rho_b$, where ρ_f is the free charge density and ρ_b bound charge density due two polarization. Now $\rho_b = -\vec{\nabla}.\vec{P}$, the polarization $\vec{P}$ is defined to be a vector field whose divergence yields the density of bound charges ρ_b in the material.

Differential form of Gauss's law for dielectric

$$\begin{aligned} \vec{\nabla}.\vec{E} &= \frac{\rho}{\varepsilon_0}, \\ \vec{\nabla}.\vec{E} &= \frac{\rho_f + \rho_b}{\varepsilon_0}, \\ \vec{\nabla}.\vec{E} &= \frac{1}{\varepsilon_0}(\rho_f - \vec{\nabla}.\vec{P}), \\ \vec{\nabla}.(\varepsilon_0\vec{E} + \vec{P}) &= \rho_f, && (1.27) \\ \therefore \quad \vec{\nabla}.\vec{D} &= \rho_f, && (1.28) \end{aligned}$$

where D is called the displacement vector. Therefore,

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} \tag{1.29}$$

The relation of Eq. 1.29 is called ***three electric vectors.***

Polarization is proportional to the electric field. So, $\vec{P} = \varepsilon_0 \chi \vec{E}$, where χ is called the electric susceptibility[2].

$$\begin{aligned} \vec{D} &= \varepsilon_0 \vec{E} + \varepsilon_0 \chi \vec{E} \\ \vec{D} &= \varepsilon_0 (1+\chi) \vec{E} \\ \vec{D} &= \varepsilon_0 \varepsilon_r \vec{E} \\ \vec{D} &= \varepsilon \vec{E}, \end{aligned} \tag{1.30}$$

where $\varepsilon_r = (1+\chi)$ is called the relative permittivity[3].

1.8 Examples

Example 1.1: Consider charge is distributed uniformly throughout infinitely long solid cylinder of radius R and charge density ρ. (a) Find E at a distance r from the cylinder axis when $r < R$, and (b) E at a distance r from the cylinder axis when $r > R$.

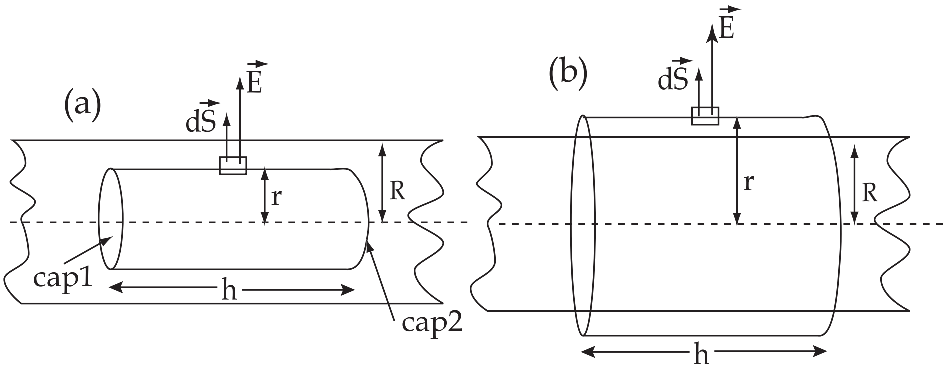


Figure 1.14: A cylindrical charge distribution of radius R. (a) Gaussian surface with radius $r < R$, (b) Gaussian surface with radius $r > R$.

Solution: (a) Consider a cylindrical Gaussian surface of radius r ($r < R$) and length h as it is shown in Fig. 1.14a. Charge enclosed by the Gaussian surface is $q = \pi r^2 h\rho$. Now using Gauss' law

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q,$$

$$\text{or} \quad \varepsilon_0 \int_{cap1} \vec{E}.d\vec{s} + \varepsilon_0 \int_{cap2} \vec{E}.d\vec{s} + \varepsilon_0 \int_{curve} \vec{E}.d\vec{s} = \pi r^2 h\rho,$$

$$\text{or} \quad \varepsilon_0 \int_{cap1} E\ ds\ \cos 90^o + \varepsilon_0 \int_{cap2} E\ ds\ \cos 90^o + \varepsilon_0 \int_{curve} E\ ds\ \cos 0^o = \pi r^2 h\rho,$$

$$\text{or} \quad 0 + 0 + \varepsilon_0 \int_{curve} E\ ds = \pi r^2 h\rho,$$

[2]Susceptibility is defined as the ability to polarize of a dielectric. The greater the electric susceptibility, the greater the ability of a material to polarize in response to the field.

[3]The permittivity is often represented by the relative permittivity ε_r which is the ratio of the permittivity ε in the dielectric and permittivity ε_0 in the vacuum. i.,e., $\varepsilon_r = \varepsilon/\varepsilon_0$

$$\text{or} \qquad \varepsilon_0 E(2\pi rh) = \pi r^2 h\rho,$$

$$\therefore \quad E = \frac{\pi r^2 h\rho}{\varepsilon_0\, 2\pi rh} = \frac{\rho r}{2\varepsilon_0}.$$

(b) Similarly consider a cylindrical Gaussian surface of radius r ($r > R$) and length h as it is shown in Fig. 1.14b. Here charge enclosed by the Gaussian surface is $q = \pi R^2 h\rho$. Now using Gauss' law

$$\varepsilon_0 \oint \vec{E}.d\vec{s} = q = \pi R^2 h\rho$$

we have by the same calculation as before

$$\varepsilon_0 \int_{cap1} \vec{E}.d\vec{s} + \varepsilon_0 \int_{cap2} \vec{E}.d\vec{s} + \varepsilon_0 \int_{curve} \vec{E}.d\vec{s} = \pi R^2 h\rho,$$

$$\text{or} \qquad \varepsilon_0 \int_{cap1} E\ ds\ \cos 90^o + \varepsilon_0 \int_{cap2} E\ ds\ \cos 90^o + \varepsilon_0 \int_{curve} E\ ds\ \cos 0^o = \pi R^2 h\rho,$$

$$\text{or} \qquad 0 + 0 + \varepsilon_0 \int_{curve} E\ ds = \pi R^2 h\rho,$$

$$\text{or} \qquad \varepsilon_0 E(2\pi rh) = \pi R^2 h\rho,$$

$$\therefore \quad E = \frac{\pi R^2 h\rho}{\varepsilon_0\, 2\pi rh} = \frac{\rho R^2}{2\varepsilon_0 r}.$$

Example 1.2: Gauss' law for gravitation is

$$\frac{1}{4\pi G}\Phi_g = \frac{1}{4\pi G}\oint \vec{g}.d\vec{s} = m,$$

where m is the enclosed mass and G is the universal gravitation constant. Derive Newton's law of gravitation from this. What is the significance of negative sign?

Solution: Consider the mass m enclosed by a spherical Gaussian surface surface of radius r.

$$\text{Then} \qquad \frac{1}{4\pi G}\Phi_g = \frac{1}{4\pi G}\oint \vec{g}.d\vec{s} = \frac{1}{4\pi G}\oint g\ ds\ \cos 180 = m,$$

$$\text{or} \quad -\frac{g}{4\pi G}\oint ds = m,$$

$$\text{or} \quad -\frac{g.4\pi r^2}{4\pi G} = m,$$

$$\text{or} \quad g = -\frac{Gm}{r^2},$$

Let a mass M is placed in the gravitational field g, which experienced a force F is found by

$$F = Mg = -\frac{GMm}{r^2},$$

which is Newton's law of gravitation. The negative sign indicates that the force is attractive.

Example 1.3: A proton orbits with a speed $v = 294\ km/s$ just outside a charged sphere of radius $r = 11.3\ cm$. Find the charge on the sphere.

Solution: Let the charge on the sphere be Q. Then the Coulomb's force between the proton (with charge q) and the sphere is

$$F_{Coul} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}.$$

Now the centripetal force

$$F_c = \frac{mv^2}{r},$$

where m is the mass of the proton. For electrostatic equilibrium

$$F_{Coul} = F_c,$$

$$\text{or} \quad \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2} = \frac{mv^2}{r},$$

$$\text{or} \quad Q = \frac{4\pi\varepsilon_0 mv^2 r}{q},$$

$$= \frac{4\pi(8.85\times10^{-12}\ C^2/N\ .m^2)(1.67\times10^{-27}\ kg)(294\times10^3\ m/s)^2(0.0113\ m)}{1.6\times10^{-19}\ C} = 1.1338\times10^{-9}\ C.$$

Example 1.4: A solid nonconducting sphere of radius R carries a nonuniform charge distribution, with charge density $\rho = \rho_s r/R$, where ρ_s is a constant and r is distance from the center of the sphere. Show that (a) the total charge of the sphere is $Q = \pi\rho_s R^3$, and (b) the electric field inside the sphere is given by $E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^4}r^2$.

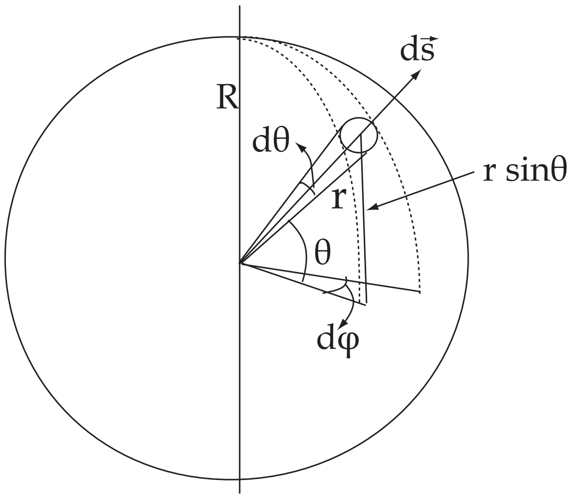


Figure 1.15: A non uniform cylindrical charge distribution with charge density $\rho = \rho_s r/R$, where ρ_s is constant.

Solution: (a) The total charge is the volume integral over the whole sphere, Therefore $Q = \int \rho\ dV$. From Fig. 1.15 we have a surface element $ds = (r\ d\theta)(r\ \sin\theta\ d\phi)$. So we have a volume element as

$$dV = (dr)(ds) = (dr)(r\ d\theta)(r\ \sin\theta\ d\phi)$$

$$\therefore \quad Q = \int \rho\ dV$$

$$= \int \left(\frac{\rho_s r}{R}\right)(dr)(r\ d\theta)(r\ \sin\theta\ d\phi)$$

$$= \frac{\rho_s}{R}\int_0^R r^3 dr \int_0^\pi \sin\theta \, d\theta \int_0^{2\pi} d\phi = \frac{\rho_s}{R}\int_0^R r^3 dr \; [-\cos\theta]_0^\pi \; [\phi]_0^{2\pi}$$

$$= \frac{4\pi\rho_s}{R}\int_0^R r^3 dr = \frac{4\pi\rho_s}{R}\left[\frac{r^4}{4}\right]_0^R$$

$$= \frac{\pi\rho_s R^4}{R} = \pi\rho_s R^3.$$

(b) Let us now consider a Gaussian surface of radius r $(r < R)$. Therefore, charge enclosed by the Gaussian surface can be calculated

$$q' = \int \rho \, dV'$$

$$= \frac{\rho_s}{R}\int_0^r r^3 dr \int_0^\pi \sin\theta \, d\theta \int_0^{2\pi} d\phi = \frac{\rho_s}{R}\int_0^r r^3 dr \; [-\cos\theta]_0^\pi \; [\phi]_0^{2\pi}$$

$$= \frac{4\pi\rho_s}{R}\int_0^r r^3 dr = \frac{4\pi\rho_s}{R}\left[\frac{r^4}{4}\right]_0^r$$

$$= \frac{\pi\rho_s r^4}{R}.$$

Now using Gauss' law

$$\varepsilon_0 \oint \vec{E}.d\vec{A} = q',$$

$$\text{or} \qquad \varepsilon_0 E(4\pi r^2) = \frac{\pi\rho_s r^4}{R},$$

$$\text{or} \qquad E = \frac{\pi\rho_s r^4}{4\pi\varepsilon_0 r^2 R} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^4}r^2.$$

Chapter 2

ELECTRIC POTENTIAL

Many electrical phenomena are associated with the transfer of large quantities of energy. For example, when a lightning flash strikes the Earth from a cloud, an energy of typically 10^8 J is released in the form of light, sound, heat, and shock wave. Where does this energy comes from and how is it stored in clouds? To understand this question, we must consider the energy associated with electrical force. The electrostatic force law is very similar to the gravitational force law:

$$F = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2} \tag{2.1}$$

$$F = G\,\frac{m_1 m_2}{r^2} \tag{2.2}$$

The difference of the forces is that the gravitational force is always attractive, whereas electrostatic force can be either attractive or repulsive. Both forces depends on the inverse square of the separation distance between the two objects. When an object moves from place to place under the gravitational force of another object, the work done is occurred by the gravitational force. Similarly when an electric charge moves from one place to another place, a work done is performed under the electrical force. The forces are conservative, because the work done does not depends on the path of the travelling. For a conservative force we could define a potential energy. The difference in potential energy ΔU of the system as the object moves from its initial to its final position is equal to the negative of the work done by the force:

$$\Delta U = U_f - U_i = -W_{if} = -\int_i^f \vec{F}.d\vec{l}, \tag{2.3}$$

where W_{if} is the work done by the force $\vec{F}$ when the object moves from place i to f. The electrostatic force is conservative, and therefore there is a potential energy associated with the configuration of a system in which electrostatic forces act.

2.1 Potential Difference and Potential

Imagine a charge q fixed at the origin of a coordinate system, as it is shown in Fig. 2.1. Let a test charge q_0 is to move from point 'a' to point 'b'. The distance of the two points be r_a and r_b, respectively. Consider $q > q_0$, therefore we have to apply a working force which will be equal to Coulomb's force but in opposite direction. The change in potential energy ΔU of the system can be calculated by Eq. 2.3

$$\Delta U = -W_{ab} = -\int_a^b \vec{F}.d\vec{l} = -\int_a^b F\ dl\ \cos 180^o = +\int_a^b F\ dl.$$

But $dl = -dr$, and we have

$$\Delta U = -W_{ab} = -\int_{r_a}^{r_b} F\ dr = -\int_{r_a}^{r_b} \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}\ dr = \frac{1}{4\pi\varepsilon_0}\ qq_0\left(\frac{1}{r_b} - \frac{1}{r_a}\right), \tag{2.4}$$

$$\text{or} \qquad \frac{\Delta U}{q_0} = \frac{-W_{ab}}{q_0} = V_b - V_a, \tag{2.5}$$

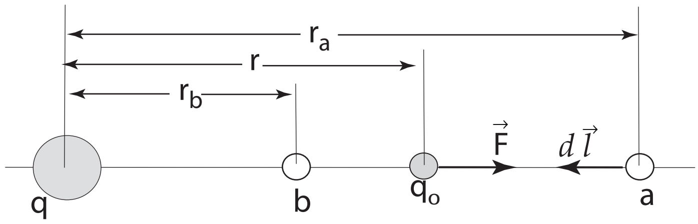


Figure 2.1:** **A test charged particle q_0 is moving from a to b under the influence of the electrostatic force $\vec{F}$ exerted by q.

$$\text{or} \qquad \frac{\Delta U}{q_0} = \Delta V. \tag{2.6}$$

We define the electric potential difference ΔV to be the electrical potential energy difference per unit test charge.

Let us consider the point 'a' is placed at infinite distance. That is $r_a = \infty$. From Eq. 2.4 we have

$$\frac{\Delta U}{q_0} = \frac{-W_{ab}}{q_0} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r_b}, \tag{2.7}$$

which gives the potential difference per unit test charge is equal to the potential at point 'b'. Therefore, taking the magnitude of Eq. 2.4 we can write in general form

$$\frac{W}{q_0} = V. \tag{2.8}$$

$$\text{If } q_0 = 1 \text{ then} \qquad W = V. \tag{2.9}$$

Therefore, the potential at any point situated in an electric field is defined as the work done in moving a unit test charge from infinite distance to that point.

2.2 Relation Between E and V

The interactions of a charge can be characterized using four different properties: electric force, electric field, electric potential energy and electric potential. Here we try to relate electric potential to electric field. The connection between V and $\vec{E}$ follows directly from the definition of potential in Eqs. 2.5 and 2.6

$$\Delta V = \frac{-W_{ab}}{q_0} = \frac{-\int_a^b \vec{F}.d\vec{l}}{q_0} = \frac{-\int_a^b q_0\vec{E}.d\vec{l}}{q_0},$$

$$\text{or} \qquad \Delta V = V_b - V_a = -\int_a^b \vec{E}.d\vec{l}. \tag{2.10}$$

Let the point 'a' to be at infinity, then $V_a = 0$ and the potential at point 'b' is given by

$$V_b = -\int_\infty^b \vec{E}.d\vec{l}. \tag{2.11}$$

2.2.1 Potential when E is uniform

Suppose we move a test charge q_0 (Fig. 2.2) through a uniform electric field $\vec{E}$ from a to b along the path acb. We try to find potential difference between a to b. For the path ac we have potential difference from Eq. 2.10,

$$V_c - V_a = -\int_a^c E\ dl\ \cos(\pi - \theta)$$

$$= E\ \cos\theta \int_a^c dl = E\ \cos\theta . ac = E\ \cos\theta . \frac{L}{\cos\theta} = EL.$$

Potential difference between c to b is

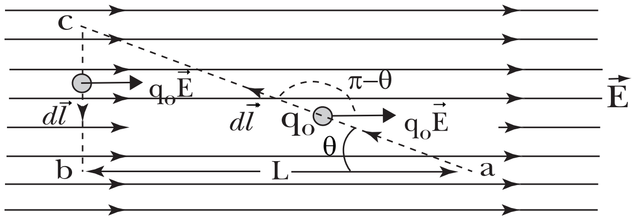


Figure 2.2: Consider a uniform electric field E. A test charge q_0 have to move from point a to point b.

$$V_b - V_c = -\int_c^b E\ dl\ \cos 90^o = 0.$$

Therefore, potential difference between a to b is given by

$$V_b - V_a = (V_c - V_a) + (V_b - V_c) = EL + 0 = EL.$$

This the same value if we follow the direct path a to b, because we know that the potential difference between two points is independent of the path.

2.2.2 Potential when E is not uniform

Let a test charge q_0 is to move from a arbitrary point 'a' to 'b'as it is shown in Fig. 2.3. The electric field exerts a force $q_0\vec{E}$ on the test charge q_0. To keep the test charge from accelerating, an external agent must be applied a force $\vec{F}$ chosen to be exactly equal to $-q_0\vec{E}$ for all position of the test

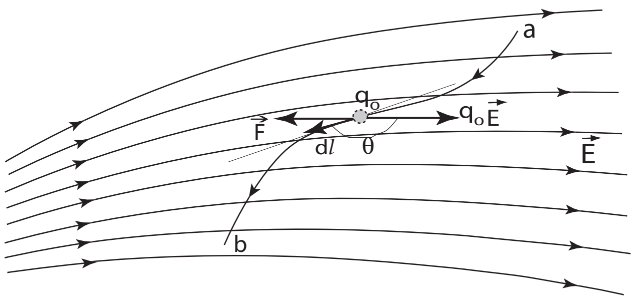


Figure 2.3: A non-uniform electric field E. A test charge q_0 has to move from point 'a' to point 'b'.

charge. To find the total work done W_{ab} be the external agent in moving the test charge from a to b is given as

$$W_{ab} = \int_a^b -q_0\vec{E}.d\vec{l}.$$

Potential difference between the points a and b

$$V_b - V_a = \frac{W_{ab}}{q_0} = -\int_a^b \vec{E}.d\vec{l} = -\int_a^b E\ dl\ \cos\theta.$$

Consider a is at infinity, then $V_0 = 0$. Hence

$$V_b = V = -\int_\infty^b \vec{E}.d\vec{l}.$$

2.3 Calculation of Potential

2.3.1 Potential due to a point charge

A point charge rendered the electric field spherically outward as shown in Fig. 2.4. Let a test charge q_0 is to move from a arbitrary point a to b. The potential difference between a to b can be calculated by using Eq. 2.10 and written as

$$V_b - V_a = -\int_a^b \vec{E}.d\vec{l} = -\int_a^b E.dl\ \cos\theta.$$

Here the angle θ between $\vec{E}$ and $d\vec{s}$ along the line ab is 180^o.

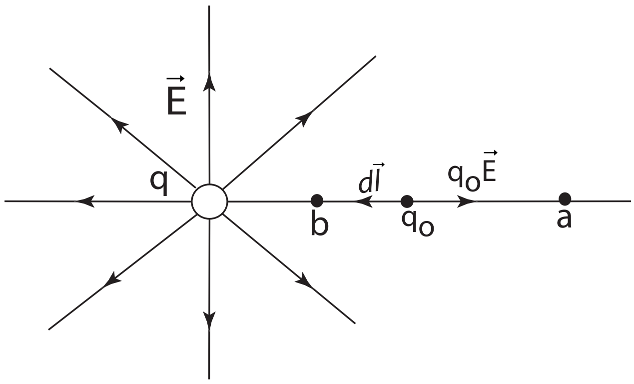


Figure 2.4: Consider a point charge and electric field E is directed radially outward. A test charge q_0 have to move from point a to point b.

$$\therefore\quad V_b - V_a = -\int_a^b E.dl\ \cos 180^o = +\int_a^b E\ dl.$$

Since the electric field is directed spherically outward,

$$\therefore\quad dl = -dr.$$

$$\therefore\quad V_b - V_a = +\int_a^b E\ dl = -\int_{r_a}^{r_b} E\ dr = -\int_{r_a}^{r_b} \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\ dr$$
$$= -\frac{q}{4\pi\varepsilon_0}\int_{r_a}^{r_b}\frac{dr}{r^2} = +\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_b} - \frac{1}{r_a}\right).$$

Let us consider A is placed at infinity, then $\frac{1}{r_a} = 0$ and $V_a = 0$, and we put $V_b = V$ and $\frac{1}{r_b} = \frac{1}{r}$. Now we have

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}. \tag{2.12}$$

Actually the expressions shown by the Eqs. 2.12 and 2.7 are the same.

2.3.2 Potential due to a collection of point charges

Suppose we have a collection of point charges q_1, q_2 ......, q_n located at various fixed points. We wish to find the potential at any arbitrary point P due to these collection of charges. We can find the potential at P by the sum of the potentials due to individual point charges.

Thus we have

$$V = V_1 + V_2 + \ldots\ldots V_n = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2} + \ldots\ldots + \frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_n}, \tag{2.13}$$

which can be written as compact form

$$V = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i}. \tag{2.14}$$

The expression indicates that q_i is the magnitude of the ith charge and r_i is the distance of the ith charge from the point P.

2.3.3 Potential due to a dipole

The configuration of two equal but opposite charges separated by a finite distance is called an electric dipole. In Fig. 2.5 two equal charges, one positive $+q$ and another negative $-q$ are separated by a distance d. The configuration is said to be an electric dipole.

The potential due to an electric dipole can be calculated in a straightforward way using Eq. 2.14. We place the origin at the center of the dipole, and we seek the potential at point P, which is located a distance r from the center of the dipole. Let the angle between r and y axis be θ. The

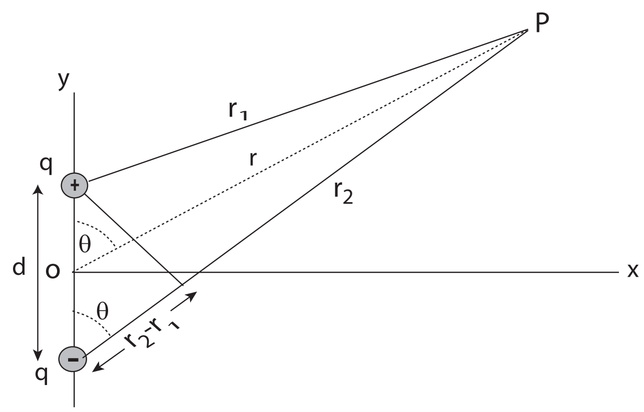


***Figure 2.5:** A configuration of electric dipole. Two equal charges $+q$ and $-q$ are separated by a distance d. O is the origin of the configuration.*

distances from the positive and negative charges to P are r_1 and r_2 respectively. Using Eq. 2.14, we find the potential at point P to be

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{+q}{r_1} + \frac{-q}{r_2}\right). \tag{2.15}$$

$$V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = \frac{q}{4\pi\varepsilon_0}\left(\frac{r_2 - r_1}{r_1 r_2}\right). \tag{2.16}$$

Equation 2.15 is the exact expression for the potential due to a dipole. However, in many applications it is considered that the point P is very far from the dipole, compared with the distance d between the charges; that is $r >> d$.

In this assumption,
$r_2 - r_1 \approx d\cos\theta$ and $r_1 r_2 \approx r^2$, Now Eq. 2.16 gives the result

$$V = \frac{1}{4\pi\varepsilon_0}\frac{qd\cos\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{p\ \cos\theta}{r^2}, \tag{2.17}$$

where $p = qd$ and is called electric dipole moment. Dipole moment p is a vector whose direction is from $-q$ to $+q$. Equation 2.17 gives the potential due to a dipole at any point in space.

Note that potential due to dipole varies as $1/r^2$ which is in contrast to the potential for a single charge which varies (see Eq. 2.12) as $1/r$. If $\theta = 90^o$, Eq. 2.17 gives the potential along the x axis as $V = 0$. Therefore, potential along the perpendicular bisector of an electric dipole is zero. For a given r, the potential varies from positive values on the positive y axis (for $\theta = 0$) to zero in the x axis (for $\theta = 90^o$) to negative values on the negative y axis (for $\theta = 180^o$). Though $V = 0$ in the x axis but actually $E \neq 0$.

We know that there is a relation between Electric field E and Electric potential V. So, if one know the potential at any point of an electric field, he can also calculate the electric field easily from that expression of potential. The relation between Electric field E and Electric potential V is

$$E = -\frac{dV}{dr}.$$

Substituting the value of V of Eq. 2.17 we have

$$E = -\frac{d}{dr}\left(\frac{1}{4\pi\varepsilon_0}\frac{p\ \cos\theta}{r^2}\right),$$

$$or \qquad E = -\left(\frac{p\ \cos\theta}{4\pi\varepsilon_0}\right)\frac{d}{dr}(r^{-2}),$$

$$or \qquad E = \frac{p\ \cos\theta}{2\pi\varepsilon_0}\ \frac{1}{r^3}. \tag{2.18}$$

Note: We see from Eqs. 2.17 and 2.18, for an electric dipole, $V \propto \frac{1}{r^2}$ and $E \propto \frac{1}{r^3}$.

2.3.4 Potential due to a quadrupole

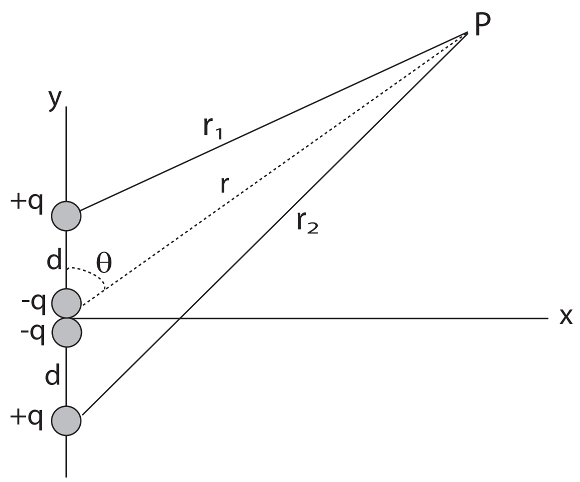


Figure 2.6: A configuration of electric quadrupole.

When two electric dipole are placed closely like as shown in Fig. 2.6, the configuration is called an *electric quadrupole*. We try to find electric potential at point P due to quadrupole. Let we choose the origin at the center of the quadrupole. The potential at P due to this quadrupole is

$$V(r) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_1} - \frac{2q}{r} + \frac{q}{r_2}\right) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} + \frac{1}{r_2} - \frac{2}{r}\right). \tag{2.19}$$

Using cosine law, we can write with respect to Fig. 2.6,

$$r_1^2 = r^2 + d^2 - 2rd\cos\theta$$

$$\text{Hence,} \qquad \frac{1}{r_1} = \frac{1}{(r^2 + d^2 - 2rd\cos\theta)^{1/2}} = \frac{1}{r}\left[1 + \left(\frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right)\right]^{-1/2}.$$

Since $r >> d$ we can expand by Binomial theorem, as,

$$(1+x)^{-1/2} = 1 - \frac{x}{2} + \frac{3x^2}{8} \cdots\cdots\cdots$$

$$\text{Hence,} \qquad \frac{1}{r_1} = \frac{1}{r}\left[1 - \frac{1}{2}\left(\frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right) + \frac{3}{8}\left(\frac{d^2}{r^2} - \frac{2d}{r}\cos\theta\right)^2 \cdots\cdots\cdots\right]$$

Neglecting higher order terms we have

$$\frac{1}{r_1} = \frac{1}{r}\left[1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) + \frac{d}{r}\cos\theta\right],$$

$$\text{and similarly} \qquad \frac{1}{r_2} = \frac{1}{r}\left[1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) - \frac{d}{r}\cos\theta\right].$$

After substituting these values in Eq. 2.19, we find

$$V(r) = \frac{q}{4\pi\varepsilon_0}\frac{1}{r}\left[1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) + \frac{d}{r}\cos\theta + 1 + \frac{d^2}{2r^2}\left(3\cos^2\theta - 1\right) - \frac{d}{r}\cos\theta - 2\right],$$

$$\text{or} \qquad V(r) = \frac{qd^2}{4\pi\varepsilon_0 r^3}\left(3\cos^2\theta - 1\right) \tag{2.20}$$

The product $2qd^2$ is called the electric quadrupole moment and can be denoted by Q, then potential due to electric quadrupole is

$$V(r) = \frac{Q}{8\pi\varepsilon_0 r^3}\left(3\cos^2\theta - 1\right) \tag{2.21}$$

Note that the potential due to quadrupole varies as $1/r^3$, whereas the potential due to dipole varies as $1/r^2$ and the potential for a single charge varies as $1/r$.

Note: Potential	*Note: Electric field*
$V \propto \frac{1}{r}$, for single charge,	$E \propto \frac{1}{r^2}$, for single charge,
$V \propto \frac{1}{r^2}$, for dipole,	$E \propto \frac{1}{r^3}$, for dipole,
$V \propto \frac{1}{r^3}$, for quadropole.	$E \propto \frac{1}{r^4}$, for quadropole.

2.4 A Dipole in an Electric Field

Figure 2.7 shows an electric dipole placed in a uniform electric field $\vec{E}$. The dipole axis makes an angle θ with the field direction E. Therefore, the charge $+q$ will experience a force $F = qE$ along the direction of field E and the charge $-q$ will experience the same force $F = qE$ but opposite to the direction of field E, as it is shown in Fig. 2.7.

These two equal and opposite forces produce a torque due to the couple of moment and is given by

$$\tau = qE \times d \sin\theta,$$

$$\text{or} \quad \tau = qd\, E \sin\theta. \tag{2.22}$$

Magnitude qd is defined by dipole moment and denoted by $p = qd$. Therefore, from Eq. 2.22 we have,

$$\tau = pE\ \sin\theta. \tag{2.23}$$

In vector notation this torque can be written as

$$\vec{\tau} = \vec{p} \times \vec{E}. \tag{2.24}$$

This torque tends to align the dipole to the direction of E. *Therefore, if a dipole placed in an external electric field experiences a torque which tends to align it with the direction of applied field.*

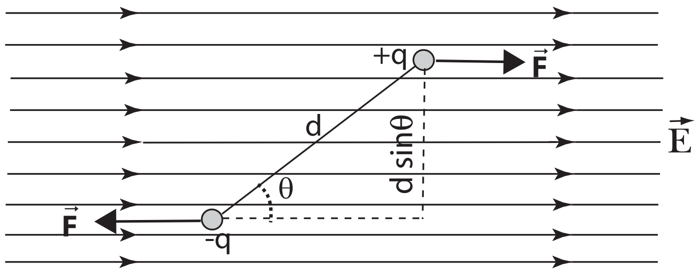


Figure 2.7: Electric dipole in a uniform electric field.

So to keep in position the dipole in the field, an external force equal to qE must be applied to it which can do work. This work done is stored as potential energy U in the system. Let the external force makes an angle $d\theta$ against the torque, then the work done by the force

$$dW = pE\ \sin\theta\ d\theta.$$

Therefore, the work done in rotating the dipole with making an angle θ which is stored as potential energy

$$U = W = \int dW = pE \int \sin\theta\ d\theta,$$

$$\text{or} \quad U = -pE\cos\theta, \tag{2.25}$$

$$\text{or} \quad U = -\vec{p}.\vec{E}. \tag{2.26}$$

Let $\theta = 0$, then from Eq. 2.25 we have

$$U = -pE.$$

Therefore, to keep p and E in parallel, the work done is minimum and stored potential energy is also minimum. Let $\theta = 90$, then from Eq. 2.25 we have potential energy $U = 0$. Let $\theta = 180$, then from Eq. 2.25 we have potential energy $U = pE$, which gives the maximum value.

Chapter 3

MAGNETIC FIELD DUE TO CURRENT

In this chapter we shall study the magnetic field produced by moving charges. We know, usually, a large number of charges move when we say a current is flowing in a wire. We shall investigate in this chapter the magnetic field produced by some simple current distribution say straight wires, circular loops. We shall also see the characteristics of produced field, lines of field and direction of the force lines as well as the amount of magnetic field related to the amount of current.

3.1 Magnetic Field Produced by Current

In 1820 Hans Christian Oersted demonstrated an experiment and observed when a compass is placed near a straight wire carrying a current, like shown in Fig. 3.1, the compass needle aligns so that it is tangent to a circle drawn around the wire. Oersted's discovery provided the first link between electricity and magnetism.

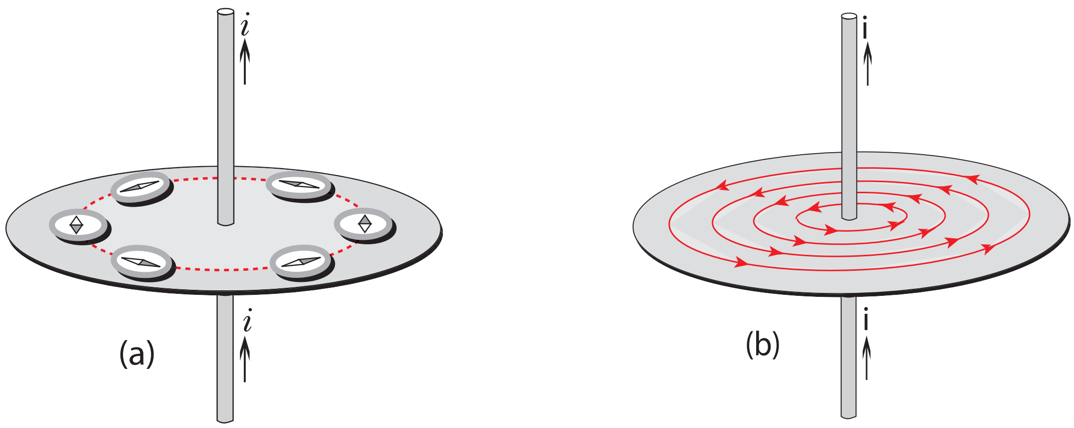


Figure 3.1: (a) Magnetic field produced by current is shown by Oersted's experiment. The direction of the compass is always perpendicular to the direction of the current in the wire. (b) The lines of magnetic field are concentric circles produced by a long straight wire of carrying current i.

3.2 Magnetic Field Due to a Moving Charge

To discuss the physics of produced magnetic field, first we wish to introduce how a single moving charge produces a magnetic field, later we will see why this approach is not practical and why it is easier to produce magnetic field in the laboratory by using moving charges in the form currents in wires.

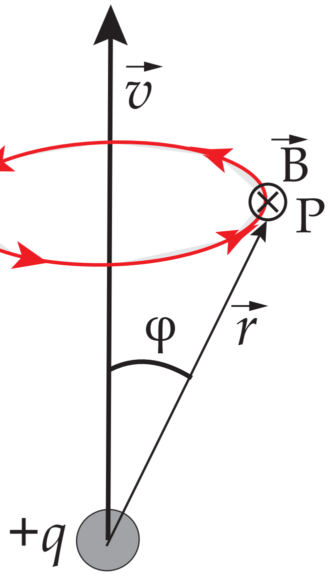


***Figure 3.2:** The magnetic field at point P due to a moving charge. $\vec{B}$ is directed perpendicularly to the page (going away from the eye sight).*

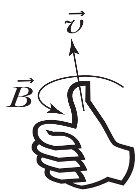
Figure 3.2 shows a moving charge $+q$ sets a magnetic field $\vec{B}$. We can also observe the properties of charge and magnetic field which determine magnitude of B. The properties are

1. The field strength is directly proportional to the magnitude of the velocity v and also to the charge q.
2. If $\vec{v}$ reverses direction or q changes sign, the direction of $\vec{B}$ is reversed.
3. The direction of $\vec{B}$ is tangent to circles drawn on planes perpendicular to the direction of $\vec{v}$, with the direction of $\vec{B}$ determined by the right-hand rule[1], and on any given circle the magnitude of $\vec{B}$ is the same at all points.
4. The magnitude of $\vec{B}$ observed at any point P depends on the distance $\vec{r}$ from the charge q. The magnitude of $B \propto 1/r^2$.
5. The magnitude of $\vec{B}$ depends on $\sin\varphi$, where φ is the angle between $\vec{v}$ and $\vec{r}$. If $\varphi = 0$, then $\sin\varphi = 0$. This means the field is zero along the direction of $\vec{v}$. Again if $\varphi = 90^o$, then $\sin\varphi = 1$. This means the field is maximum if $\vec{r}$ is perpendicular to the direction of $\vec{v}$. In Fig. 3.2 the direction of $\vec{B}$ at point P is perpendicular to the plane determined by the vectors $\vec{v}$ and $\vec{r}$. From the above observations we find that the magnitude of $\vec{B}$ is directly proportional to q, $\vec{v}$ and $\sin\varphi$, and inversely proportional to r^2.

Finally we can write

$$B \propto \frac{qv\sin\varphi}{r^2},$$

[1] point your thump in the direction of $\vec{v}$, and your fingers will curl in the direction of $\vec{B}$.



$$\text{or} \qquad B = K\frac{qv\sin\varphi}{r^2}, \tag{3.1}$$

where K is a constant of proportionality.

It indicates that φ is the angle between the vectors $\vec{v}$ and $\vec{r}$, and $\vec{B}$ is directed perpendicularly to the plane containing $\vec{v}$ and $\vec{r}$, provided that the direction of B follows the right-hand rule.

The proportionality constant K is defined in SI unit $K = \frac{\mu_0}{4\pi} = 10^{-7}$ tesla.meter/ampere (T.m/A). Hence

$$\mu_0 = 4\pi \times 10^{-7} T.m/A,$$

where μ_0 is called permeability constant, here simply magnetic constant. The magnetic constant μ_0 plays a role in calculating magnetic field to that of the electric constant ε_0 in calculating electric field. The two constant are related through the speed of light as $c = (\varepsilon_0\mu_0)^{-1/2}$. The magnitude of $\vec{B}$ can be written as

$$B = \frac{\mu_0}{4\pi}\frac{qv\sin\varphi}{r^2}. \tag{3.2}$$

In vector form Eq. 3.2 can be written as

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\vec{r}}{r^3}. \tag{3.3}$$

where $\hat{r}$ is the unit vector along $\vec{r}$. Equations 3.2 or 3.3 are not practically useful because of difficulties of measuring of q. In practice it is easy to measure i instead of q.

3.3 The Biot-Savart Law

The relations of Eqs. 3.2 or 3.3 are not usually useful. In electrostatics we observed that the electric field produced by charges whose location did not change. Here we are also interested for magnetostatics; the production of steady magnetic fields by charges whose motion does not change.

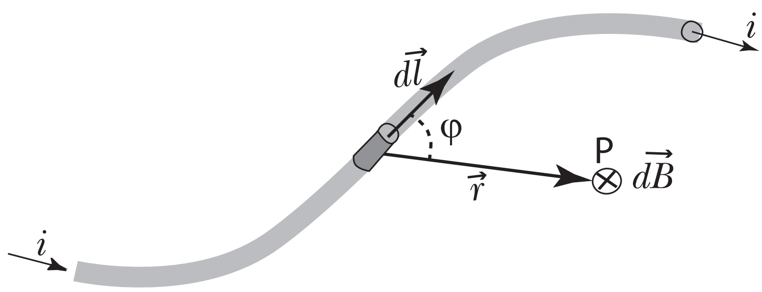


Figure 3.3: The magnetic field $d\vec{B}$ at point P produced by an element $d\vec{l}$ of an wire carrying current i. $d\vec{B}$ is directed perpendicularly to the plane formed by $d\vec{l}$ and $\vec{r}$.

In laboratory we produce magnetic fields using current carrying wires rather than the motion of individual charges, as it is shown in Fig. 3.3. If here we consider a steady-state current in the wire, it is obvious that a certain amount of charge per second passes through any cross sectional area at any instant of time which may create a steady magnetic field at any point near the wire.

Let the current element dl contains an amount of charge dq in the time interval dt, and the steady-state velocity of a charge in the wire be v. Now the relations for dB produced by the moving charge dq at point P with distance r from the location of charge can be written as

$$\begin{aligned} dB &\propto dq \\ dB &\propto v \\ dB &\propto \sin\varphi \\ dB &\propto \frac{1}{r^2} \end{aligned}$$

From the above relations let's find an equation for dB at point P as

$$dB = \frac{\mu_0}{4\pi}\frac{dq\, v\sin\varphi}{r^2}. \tag{3.4}$$

Notice that the Eq. 3.4 is similat to Eq. 3.2.

Now we can write $v = dl/dt$, hence

$$dq\, v = dq\frac{dl}{dt} = \frac{dq}{dt}dl = i\, dl.$$

Substituting this in Eq. 3.4, we obtain the Biot-Savart law in scalar formation can be rewritten by as follows

$$dB = \frac{\mu_0}{4\pi}\frac{i\, dl\sin\varphi}{r^2}, \tag{3.5}$$

where φ is the angle between the vectors dl and r.

The expression of Eq. 3.5 is known as the ***Biot-Savart law***. The direction of dB is perpendicular to the plane containing the vectors dl and r. The direction of dB follows the so called right-hand rule. The symbol $\otimes$ shown in Fig. 3.3 indicates the direction of $d\vec{B}$ which is perpendicular to the page going away from the observer.

The Biot-Savart law in vector form

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{i\, d\vec{l}\times\hat{r}}{r^2} = \frac{\mu_0}{4\pi}\frac{i\, d\vec{l}\times\vec{r}}{r^3}. \tag{3.6}$$

To calculate the total field at any point P due to the entire current distribution, one have to integrate over all current element $i\, dl$. Therefore,

$$\text{or} \qquad B = \int dB = \frac{\mu_0}{4\pi}\int\frac{i\, dl\sin\varphi}{r^2}. \tag{3.7}$$

3.4 Applications of Biot-Savart law

We now consider how to apply the Biot-Savart law to calculate the magnetic fields of some current-carrying wires of different shapes.

3.4.1 A straight wire segment

Figure 3.4 shows a current segment of length L carrying current i. Consider the wire segment lies along the y axis and current is flowing along $+y$ direction. We have to calculate magnetic field B at point P, a distance d from the current segment. Let us consider a small current element $d\vec{l}$ and suppose d lies along the x axis. The distance between dl and point P is r. The contributed magnetic field $d\vec{B}$ will produce at point P due to element dl. The direction of $d\vec{B}$ involves the vector cross product $d\vec{l}\times\vec{r}$ and should be followed right-hand rule. Therefore, in Fig. 3.4 $d\vec{B}$ can be indicated perpendicularly on the xy plane through $-z$ direction.

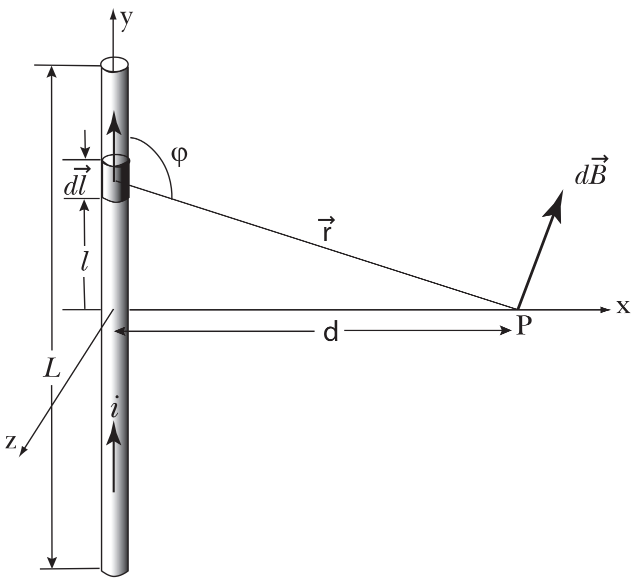


Figure 3.4: The magnetic field $d\vec{B}$ at point P produced by a current segment L of carrying current i. $d\vec{B}$ is directed perpendicularly on the xy plane through $-z$ direction.

Using Biot-Savart law the magnitude of $d\vec{B}$ at pint P can be calculated by the formula

$$dB = \frac{\mu_0}{4\pi}\frac{i\,dl\sin\varphi}{r^2}.$$

From figure we can write

$$r = \sqrt{l^2+d^2}$$

$$\text{and} \qquad \sin\varphi = \sin(\pi-\varphi) = \frac{d}{r} = \frac{d}{\sqrt{l^2+d^2}}.$$

Substituting the above values of r and $\sin\varphi$ we have from the Biot-Savart law

$$dB = \frac{\mu_0 id}{4\pi}\frac{dl}{(l^2+d^2)^{3/2}}. \tag{3.8}$$

Using integration limit $-L/2$ to $+L/2$, we find the total field;

$$B = \int dB = \frac{\mu_0 id}{4\pi}\int_{l=-L/2}^{l=+L/2}\frac{dl}{(l^2+d^2)^{3/2}},$$

Therefore, the integration is follows as[2]

$$B = \int dB = \frac{\mu_0 id}{4\pi}\left[\frac{l}{d^2(l^2+d^2)^{1/2}}\right]_{l=-L/2}^{l=+L/2},$$

$$\text{or} \qquad B = \int dB = \frac{\mu_0 i}{4\pi d}\frac{L}{(L^2/4+d^2)^{1/2}}. \tag{3.9}$$

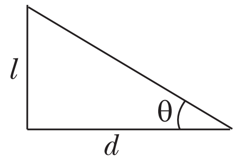
[2] $\int\frac{dx}{\sqrt{(x^2+a^2)^3}} = \frac{x}{a^2\sqrt{x^2+a^2}}$

What happened if the wire is to be considered infinitely:

Then Eq. 3.8 can be written using integration limit $l = -\infty$ to $+\infty$

$$B = \int dB = \frac{\mu_0 i}{4\pi} \int_{l=-\infty}^{l=+\infty} \frac{d}{(l^2+d^2)^{3/2}} dl. \tag{3.10}$$

To evaluate the integral let us take a trigonometric technic, suppose



$$l = d \tan\theta,$$

$$\therefore dl = d \sec^2\theta \, d\theta,$$

when $l = +\infty$ then $\theta = +\pi/2$ and when $l = -\infty$ then $\theta = -\pi/2$.
Now from Eq. 3.10 we obtain,

$$B = \frac{\mu_0 i}{4\pi} \int_{\theta=-\pi/2}^{\theta=+\pi/2} \frac{d^2 \sec^2\theta d\theta}{\{d^2(\tan^2\theta+1)\}^{3/2}},$$

$$\text{or} \qquad B = \frac{\mu_0 i}{4\pi d} \int_{-\pi/2}^{+\pi/2} \frac{1}{\sec\theta} d\theta,$$

$$\text{or} \qquad B = \frac{\mu_0 i}{4\pi d} \int_{-\pi/2}^{+\pi/2} \cos\theta d\theta = \frac{\mu_0 i}{4\pi d} [\sin\theta]_{-\pi/2}^{+\pi/2},$$

$$\text{or} \qquad B = \frac{\mu_0 i}{4\pi d} \{\sin(+\pi/2) - \sin(-\pi/2)\} = \frac{\mu_0 i}{4\pi d} (\sin\pi/2 + \sin\pi/2),$$

$$\text{or} \qquad B = \frac{\mu_0 i}{2\pi d}. \tag{3.11}$$

3.4.2 A circular current loop

Figure 3.5 shows a circular loop of radius R carrying current i. Let us calculate $\vec{B}$ at a point P on the axis a distance x from the center of the loop. The Biot-Savart law

$$dB = \frac{\mu_0}{4\pi} \frac{i \, dl \sin\varphi}{r^2}.$$

In this problem, the angle φ lies between the vectors $d\vec{l}$ and $\vec{r}$ is 90^o. Hence the Biot-Savart law stands here

$$dB = \frac{\mu_0 i}{4\pi} \frac{dl}{r^2}. \tag{3.12}$$

The vector $d\vec{B}$ is at right angles to the plane formed by $d\vec{l}$ and $\vec{r}$.

Let us resolve $d\vec{B}$ into components; one $dB\cos\alpha$ along the axis of the loop and another $dB\sin\alpha$ at right angles to the axis. Let us consider an analogous current element $d\vec{l}$ lies at the opposite position of the loop as it is considered in the previous account. Due to this argument the previous component $dB\sin\alpha$ must be cancelled by an equal and oppositely directed amount of $dB\sin\alpha$. Thus the sum of all $dB\sin\alpha$ for complete loop acting perpendicularly on the axis is zero. Only $dB\cos\alpha$ contributes to the total magnetic field $\vec{B}$ at point P.

Therefore, the the total magnetic field at point P arises

$$B = \int dB \, \cos\alpha. \tag{3.13}$$

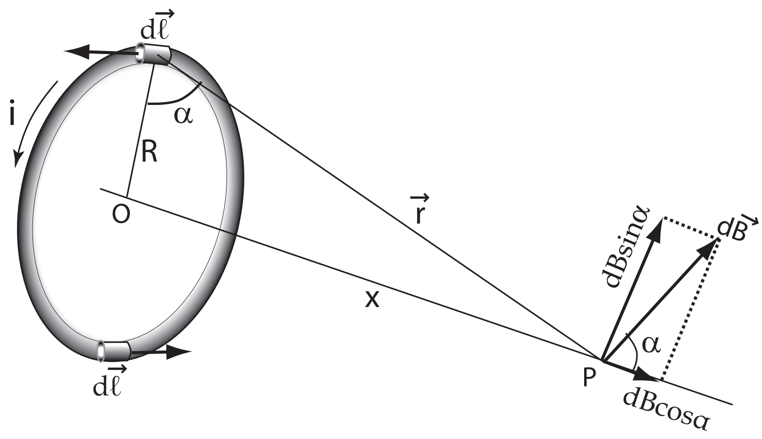


***Figure 3.5:** The magnetic field $d\vec{B}$ at point P produced by an element $d\vec{l}$ of an wire carrying current i. $d\vec{B}$ is directed perpendicularly to the plane formed by $d\vec{l}$ and $\vec{r}$.*

Now from Eqs. 3.12, 3.13 we have

$$B = \frac{\mu_0 i}{4\pi}\int \frac{\cos\alpha\, dl}{r^2}. \tag{3.14}$$

We see in Fig. 3.5 that α and r are not independent of each other. The relationships are

$$r = \sqrt{R^2 + x^2}$$

$$\text{and} \qquad \cos\alpha = \frac{R}{r} = \frac{R}{\sqrt{R^2+x^2}}.$$

Substituting these values into Eq. 3.14 we obtain

$$B = \frac{\mu_0 i R}{4\pi(R^2+x^2)^{3/2}}\int dl,$$

$$\text{or} \qquad B = \frac{\mu_0 i R}{4\pi(R^2+x^2)^{3/2}} \times 2\pi R,$$

$$\text{or} \qquad B = \frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}}, \tag{3.15}$$

note that $\int dl$ is simply the circumference of the loop $(= 2\pi R)$. If one wants to calculate B at the center of the loop then he has to consider $r = R$ everywhere. The Biot-Savart law gives that

$$B = \int dB = \frac{\mu_0 i}{4\pi r^2}\int dl = \frac{\mu_0 i}{4\pi R^2} \times 2\pi R,$$

$$\text{or} \qquad B = \frac{\mu_0 i}{2R}. \tag{3.16}$$

The result could be found also from Eq. 3.15 by putting $x = 0$.

If $x >> R$ i,e, very far from the current loop B gives from Eq. 3.15 by putting $R \approx 0$

$$B = \frac{\mu_0 i R^2}{2x^3}. \tag{3.17}$$

This dependence of the field on the inverse cube of the distance reminds as of the electric field on an electric dipole (discussed in Section 2.3 on page 20). *So it is often convenient to consider a current loop of wire to be a magnetic dipole. Just as the electric behavior of many molecules can be characterized in terms of electric dipole moment, so also the magnetic behavior of atoms can be described in terms of their magnetic dipole moment. In the case of atoms, the current loop is due to the circulation of electrons about the nucleus.*

3.5 Magnetic Field in a Solenoid

A solenoid is a helical wire wound on a cylindrical core. The wire windings carry a current i. The use of a solenoid is to get a nearly uniform magnetic field in a system produced by current. A solenoid is a long cylindrical device compering its breadth or radius. So a large number of windings are needed to form a solenoid and n number of windings are considered per unit length. Inside the solenoid it is considered that the field distribution has the axial symmetry and the magnetic force lines are parallel to the axis of the cylindrical solenoid. How to produce the magnetic field in a solenoid is shown below by the Figs. 3.6 and 3.7.

3.5.1 Loosely wire wound solenoid

Figure 3.6 shows a cross section of a solenoid having loosely wire wound, cutting through its length. The symbol $\otimes$ shown in the figure indicates the direction of current i, which is perpendicular to the page going away from the observer and the symbol $\odot$[3] indicates the direction of current perpendicular to the page coming to the observer. The circular behavior of the magnetic field close

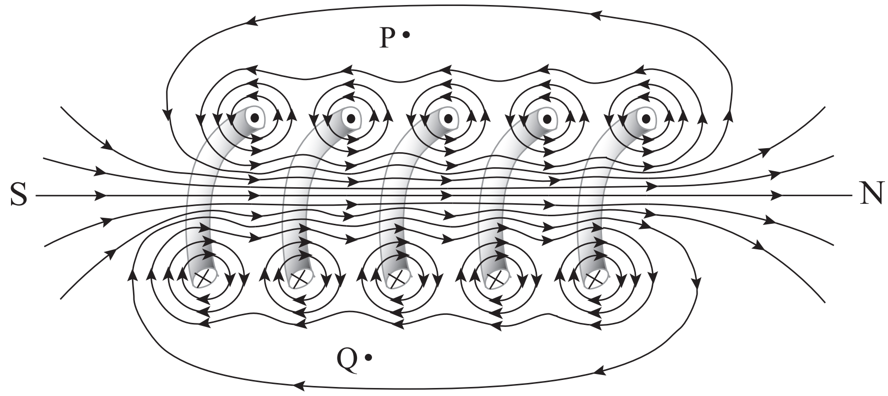


Figure 3.6: Formation of magnetic field in a loosely wound current carrying solenoid. Cross section along the length of the solenoid.

to a single wire is same as that of single straight wire. When two circular force lines regarding two adjacent wires interact each other, they combine together into a single line and exists around the adjacent wires. Thus the total field of the entire solenoid is the vector sum of the fields set up by all the turns. Lines of B tend to form parallel to each other in the interior points of the solenoid. For an ideal solenoid, the field becomes uniform and parallel to the axis and at exterior points, such as P and Q the magnetic field is considered negligible comparing to the interior points. However, interior lines of force of a loosely wire wound solenoid may not be completely parallel as it is shown in Fig. 3.6 and field intensity may also not be uniform at each point.

3.5.2 Closely wire wound solenoid

Figure 3.7 shows a cross section of a solenoid having closely wire wound, cutting through its length. To get an ideal solenoid, the windings of the wire should be very close-packed helix; so that the force lines in interior of the solenoid are almost be parallel to each other. In this case the field intensity along the axial direction is almost be uniform. Taking the external field to be zero if its length is much greater that its radius and if we consider only external points such as P and Q. The solenoid shown in Fig. 3.7 is also not an ideal because its length is not much greater than its radius.

[3] the symbol $\otimes$ means an observer is just looking the back view and the symbol $\odot$ just looking the front view of an arrow.

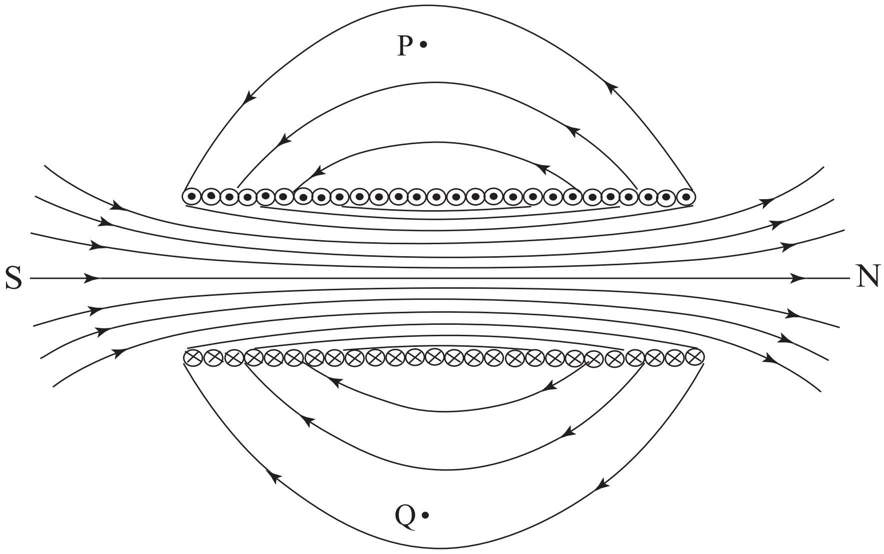


Figure 3.7: Formation of magnetic field in a closely wound current carrying solenoid. Cross section along the length of the solenoid.

A relatively simple device capable of producing a field that is approximately uniform is the parallel plate capacitor if the plate separation is small compared to the dimensions of the plate.

3.6 Ampere's Law

Coulomb's law can use to calculate the electric field associated with any distribution of electric charges. In Chapter 1, however, we have seen that Gauss's law is more suitable to solve a certain class of problems, those containing a high degree of symmetry of field distribution. We consider Gauss's formula is more basic formula than that of Coulomb's formula and Gauss's law is one of the four fundamental of Maxwell's equations of electromagnetism. The situation is similar to magnetism, where using the Biot-Savart law, one can calculate the magnetic field of any distribution of current. But to calculate magnetic field $\vec{B}$, Ampere's law is convenient where the field has symmetry like current in a long straight wire described in the subsection 3.4.1. Ampere's law is considered more fundamental than the Biot-Savart law and leads to another of the Maxwell's law of electromagnetism.

Ampere's law is written by the formula

$$\oint \vec{B}.d\vec{l} = \mu_0 i. \tag{3.18}$$

In Chapter 1, using Gauss's law the integration were performed over an imaginary closed surface called Gaussian surface. In using Ampere's law similarly the integration is performed over imaginary closed curve called Amperian loop.

Here i is the net current enclosed by the Amperian loop. The left side of Eq. 3.18, the $d\vec{l}$ is small segment of imaginary loop. The symbol $\oint$ indicates the integration all around the closed loop. The dot product of $\vec{B}$ and $d\vec{l}$ gives the scalar quantity $Bdl\cos\theta$. Where θ is angle between $\vec{B}$ and $d\vec{l}$ at any segment.

Therefore, Ampere's law can be rewritten as

$$\oint \vec{B}.d\vec{l} = \oint Bdl\cos\theta = \mu_0 i. \tag{3.19}$$

The magnetic field $\vec{B}$ at any point is the net effect of the currents in all wires. In Fig. 3.6(a) and 3.6(b) the net currents are $i = i_1 + I_2 + i_3$ and $i = i_1 - I_2 + i_3$, respectively.

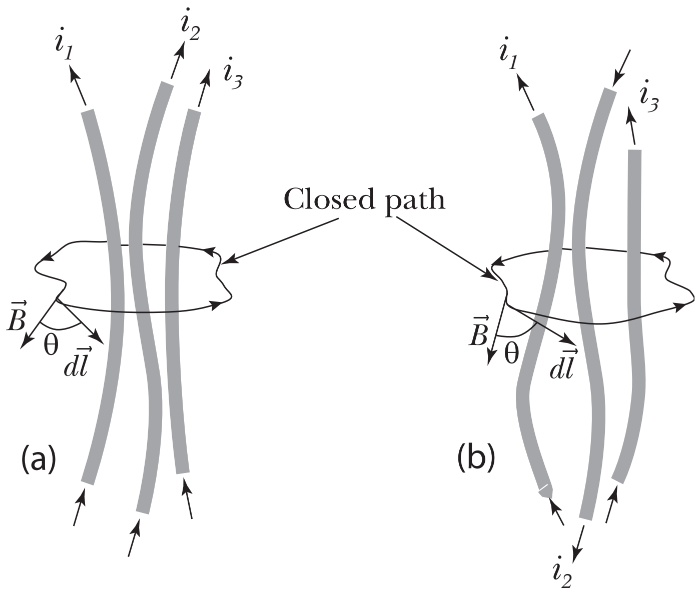


Figure 3.8: ***In Applying Ampere's law, the integration is performed around a closed contour that enclosed the net current in wires. (a) Net current enclosed by the path is $i = i_1 + I_2 + i_3$. (b) Net current enclosed by the path is $i = i_1 - I_2 + i_3$.***

3.7 Application of Ampere's Law

As it is mentioned before that Ampere's law is convenient to calculate magnetic field $\vec{B}$, where the fields have symmetric configurations. Application of Ampere's law in some more general cases are described below.

3.7.1 A long straight wire

B for an external point of a long straight current carrying wire:

In the subsection 3.4.1, we have already calculated the magnetic field B for a straight wire using Biot-Savart law. Let us now use Ampere's law to find B in the same case which is more easier process to calculate it. Figure 3.9 shows a top view of the cross section of a long current carrying wire. We can choose a circle of radius r as Amperian loop for closed path integration. The directions of $\vec{B}$ and $d\vec{l}$ at any point on the Amperian loop will be along the tangent drawn at that point of the loop. B for an external point, the radius r of the loop must be greater than the radius of the wire as

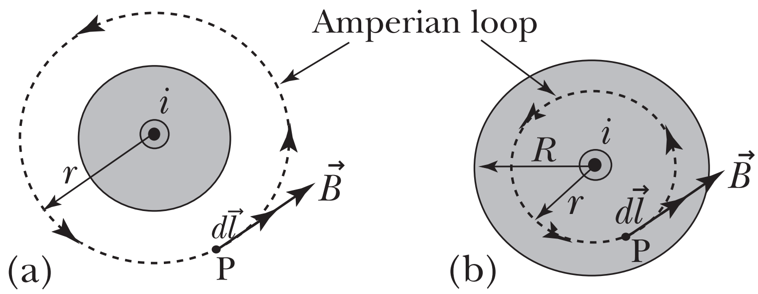


Figure 3.9: ***$\vec{B}$ for a long straight current carrying wire. (a) $\vec{B}$ for an external point, (b) $\vec{B}$ for an internal point.***

it is shown in Fig. 3.9(a). We know Ampere's law

$$\oint \vec{B}.d\vec{l} = \oint B dl \cos\theta = \mu_0 i.$$

The magnitude of $\vec{B}$ is constant around the circular path. The value of θ is zero, because $\vec{B}$ and $d\vec{l}$ have the same direction. Hence from above formula we have

$$B \oint dl = \mu_0 i,$$

$\oint dl$ simply gives the circumference of the circle of radius r. Hence

$$B(2\pi r) = \mu_0 i,$$

$$\text{or} \qquad B = \frac{\mu_0 i}{2\pi r}. \tag{3.20}$$

The result of Eq. 3.20 is identical with Eq. 3.11, a result we obtained using Biot-Savart law.

B for an internal point of a long straight current carrying wire:

We can also use Ampere's law to find magnetic field inside a wire. B for an external point, we choose the radius r of the Amperian loop less than the radius of the wire $(r < R)$ as it is shown in Fig. 3.9(b). Here first we have to know the effective current that passing through the area enclosed by the Amperian loop.

Let the effective current be i' which gives

$$i' = i\frac{\pi r^2}{\pi R^2}.$$

We have from Ampere's law

$$\oint B dl \cos\theta = B \oint dl = \mu_0 i' = \mu_0 i \frac{\pi r^2}{\pi R^2},$$

$$\text{or} \qquad B(2\pi r) = \mu_0 i \frac{\pi r^2}{\pi R^2},$$

$$\text{or} \qquad B = \frac{\mu_0 i r}{2\pi R^2}. \tag{3.21}$$

If we consider $r = R$, i.e. B on the surface gives the value from Eq. 3.21

$$B = \frac{\mu_0 i}{2\pi R}. \tag{3.22}$$

We see from Eqs. 3.21 and 3.22 that the value of B on the surface of the wire is greater than that of any interior point. We see from Eq. 3.21 $B \propto r$, and from Eq. 3.20 $B \propto r^{-1}$. Therefore, starting from the center of the wire, B at any point increases upto the surface and again decreases from the surface of the wire.

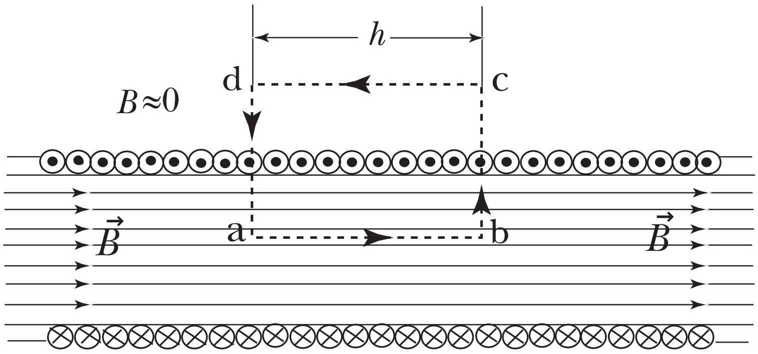


Figure 3.10:** $\vec{B}$ **for an ideal solenoid. Doted rectangle abcd is the Amperian loop.

3.7.2 A solenoid

Figure 3.10 shows an example of ideal solenoid. In this analysis we assume the magnetic field is parallel to the axis of this ideal solenoid and constant in magnitude along the axis. It also suggest that for an ideal solenoid the field is uniform at any point in the interior point. We have to calculate B for an interior point of the solenoid. For our convenient let us choose an Amperian loop of rectangular shape, where the direction of integration would be perform in the path *abcda*.

Applying Ampere's law we have

$$\oint \vec{B}.d\vec{l} = \mu_0 i'. \tag{3.23}$$

The total integration $\oint \vec{B}.d\vec{l}$ can be calculated as the sum of integration of four individual segments along the path *abcda*. Which follows

$$\oint \vec{B}.d\vec{l} = \int_a^b Bdl\cos 0^o + \int_b^c Bdl\cos 90^o + \int_c^d (0)dl\cos 180^o + \int_d^a Bdl\cos 90^o,$$

$$\text{or} \qquad \oint \vec{B}.d\vec{l} = Bh + 0 + 0 + 0 = Bh, \tag{3.24}$$

the integration for the segment c to d where the value of B outside the solenoid is ideally zero.

Now we look the effective current i' enclosed by the Amperian loop. Let the number of turns per unit length of the solenoid be n and the current carrying for a single loop be i. Then the effective current is given by

$$i' = nhi. \tag{3.25}$$

Now from Eqs. 3.23, 3.24 and 3.25 we have

$$Bh = \mu_0 nhi,$$

$$\text{or} \qquad B = \mu_0 ni. \tag{3.26}$$

Equation 3.26 concludes that the magnetic field inside an ideal solenoid is uniform over its cross section, and just knowing the number of turns per unit length and amount of current in a winding one can calculate magnetic field at any point inside it.

3.7.3 A toroid

Figure 3.11 shows a cross section of an electrical device named toroid. A toroid may be visualized to be a solenoid bent into a ring shape. The lines of field $\vec{B}$ inside the toroid are formed by a number of concentric circles and have the symmetry with uniform field distribution. We have to calculate B for a point inside it. Let us consider a concentric circle of radius r as an Amperian loop. Applying

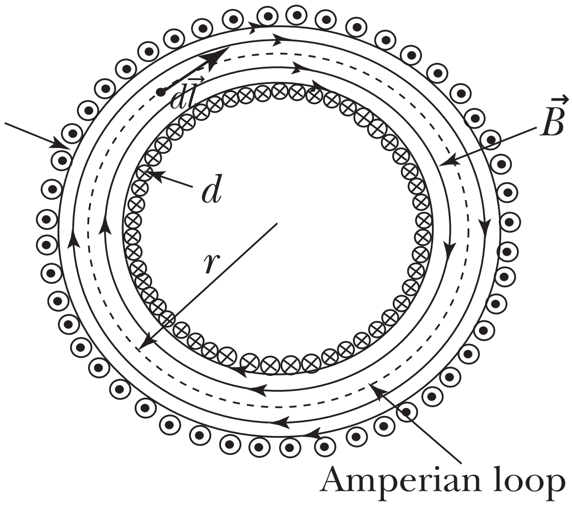


***Figure 3.11:** $\vec{B}$ for a toroid. The magnetic force lines inside the toroid are concentric circles.*

Ampere's law

$$\oint \vec{B}.d\vec{l} = \oint Bdl\cos 0^o = B\oint dl = B(2\pi r) = \mu_0 iN,$$

$$\text{or} \qquad B = \frac{\mu_0 iN}{2\pi r}, \tag{3.27}$$

where i is the current in the toroid windings and N is the total number of turns. Note that the Amperian loop encloses the total number of turns N of the toroid. Equation 3.27 discusses that B is inversely proportional to r. Therefore, we can observe that B for an ideal solenoid is constant everywhere in the interior points, in contrast that B for an ideal toroid is not constant everywhere in the interior points. One can able to show that, using Ampere's law, $B = 0$ for points outside and in the central cavity of an ideal toroid.

As we stated earlier that a toroid is a solenoid bent into the shape of a ring. Let us see Eq. 3.27 justifies our earlier statement as setting $n = N/2\pi r$ the number of turns per unit length. Then Eq. 3.27 reduces to $B = \mu_0 ni$ which has analogy of Eq. 3.26.

3.8 Examples

Example 3.1: In the Bohr model of the hydrogen atom, the electron circulates around the nucleus in a circular path of radius 5.29×10^{-11} m at a frequency f of 6.60×10^{15} Hz. What value of B is set up at the center of the orbit?

Solution: The rate of charge passes any point on the orbit gives the current at any point on the orbit and is given by

$$i = ef,$$

$$\text{or} \qquad i = (1.60\times10^{-19}C)(6.60\times10^{15}Hz) = 1.06\times10^{-3}A.$$

The magnetic field B at the center of the orbit is given by Eq. 3.16

$$B = \frac{\mu_0 i}{2R} = \frac{(4\pi\times10^{-7}T.m/A)(1.06\times10^{-3}A)}{2(5.29\times10^{-11}m)} = 12.6T.$$

Example 3.2: A solenoid 1.33 m long and 2.60 cm in diameter carries a current of 17.8 A. The magnetic field inside the solenoid is 22.4 mT. Find the length of the wire forming the solenoid.

Solution: The magnetic field inside an ideal solenoid is given by the Eq. 3.26 on page 35,

$$B = \mu_0 ni.$$

Hence the number of turns per unit length of the solenoid,

$$n = \frac{B}{\mu_0 i} = \frac{0.0224\ T}{(4\pi \times 10^{-7}\ T.m/A)(17.8\ A)} = 1.00 \times 10^3\ m^{-1}.$$

The length of the solenoid is 1.33 m. So the total number of turns of the solenoid be

$$N = nL = (1.00 \times 10^3\ m^{-1})(1.33\ m) = 1330.$$

Now radius of the solenoid i.e., each turn of the coil $r = 2.60\ cm = 0.013\ m$. Therefore total length wire which forms this solenoid is

$$= N(2\pi r) = (1330)(2\pi)(0.013\ m) \approx 109\ m.$$

Example 3.3: Figure 3.12 shows a an arrangement known as ***Helmholtz coil***. It consists of two circular coaxial coils each of N turns and radius R, separated by a distance R. They carry equal current i in the same direction. Find the field at P, mid way between the coils.

Solutions: From the Subsection 3.4.2 magnetic field on the axis of a circular loop, we can use the formula 3.15 on page 30 as

$$B = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}.$$

For magnetic field at P, here we have to introduce the turns factor N, $x = R/2$ and magnitude will

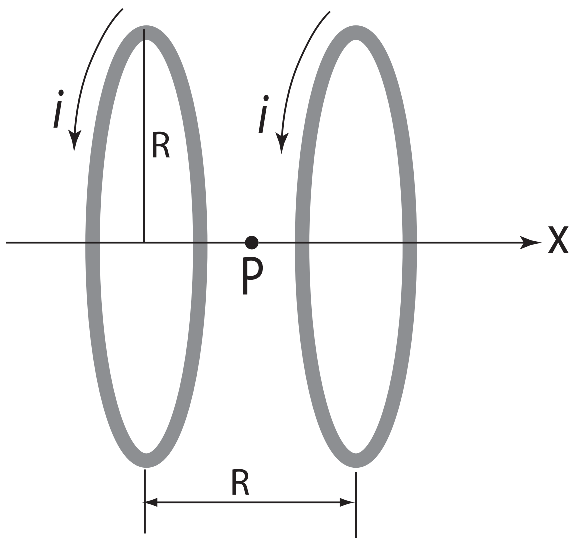


Figure 3.12: Helmholtz coil of radius R. The point P is placed in the middle of the coils on their axis.

be twice for the two coils, and we can replace the above formula as

$$B = \frac{2\mu_0 N i R^2}{2(R^2 + (R/2)^2)^{3/2}} = \frac{8\mu_0 N i}{(5)^{3/2} R}.$$

Chapter 4

FARADAY'S LAW OF INDUCTION

In the previous chapter (Chap. 3), we have studied that how a flow of current induces a magnetic field. In this chapter we shall discuss how a magnetic field induces an emf in a circuit. Faraday's law of induction gives the basic idea of electromagnetism and the equation produces by this law is one of the four Maxwell's equation. Electric and magnetic fields with change of time are related to each other. In the following sections we will relate the electric field with changing of magnetic field and demonstrate some simple experiments that can easily be done in the laboratory.

4.1 Faraday's Experiment

Michael Faraday first discovered in England in 1831 through an experiment that a magnetic field can produce an electric field. He suggested a law how a magnetic field produces current in a circuit and according to his honor this law is named as Faraday's law of induction. Figure 4.1 shows a coil

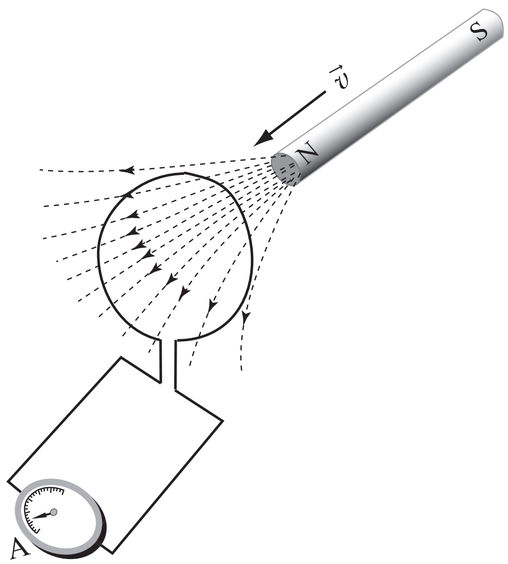


***Figure 4.1:** When the magnet is moving with respect to the coil, the ammeter 'A' shows a deflection which indicates the appearance of a current in the coil.*

of conducting wire makes a circuit connecting an ammeter in series. Normally we would expect the ammeter to show no deflection in the circuit because there is no driving source as connected to the circuit. However, if we push a bar magnet towards the coil with its north pole facing the coil, astonishingly a physical property shows there. While the magnet is moving, the ammeter deflects, showing that a current has been set up in the coil. The ammeter does not show any deflection if we hold the magnet stationary with respect to the coli. Again, if we move the magnet away from the coil, the ammeter repeats to show the deflects but in the opposite direction, which explains the current in the coil appears in the opposite direction. If we push the south pole instead of north pole in the same manner the ammeter shows also a deflection but in reversed direction as described before. The described results are also observed if we move the coil in keeping the magnet stationary.

The electromotive force that appears in the circuit is called *an induced electromotive force (emf)* and the induced *emf* set up a current in the coil is called *an induced current*.

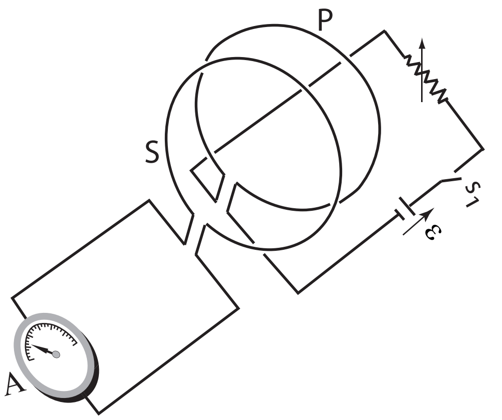


Figure 4.2: When the switch s_1 is operated in the primary coil P as on-off, the ammeter 'A' deflects immediately as the sequence of operation. The coils are not necessary to move.

Figure 4.2 shows an experimental arrangement where a magnet is not necessary to achieve a magnetic field. The two coils (one is called primary P and another is secondary S) are placed close together but at rest with respect to each other. When we close the switch s_1 in the primary coil P, the ammeter in the secondary coil S deflects immediately. Here the physical phenomenon is occurred that when we close the switch the current that flows in the primary coil P produces a magnetic field around that coil. This magnetic field cross the secondary coil S and induces an emf which makes the cause of deflection in the ammeter. When we open the switch s_1 the indicator of the ammeter shows again a movement immediately, but in the opposite direction. The experiment shows that there is an induced emf in the secondary coil whenever the current in the primary coil is changing. The common feature of these two experiment is motion or change. It is the moving magnet or the changing current that is responsible for the induced emf.

4.2 Faraday's Law

Faraday's law is a successful law of induction which can deduce the magnitude and direction of the induced emf. In practise this induced emf is very important, as for example , the current in a commercial electric generator are operated in the phenomenon of producing induced emf.

We see from Figs. 4.1 and 4.2 that an induced emf is produced in a coil whenever a number of lines of magnetic field cross that coil with changing their number of force lines. We know that the change of force lines through a cross sectional area means the change of flux in that area. Therefore, *the change of magnetic flux passing through the cross sectional area of a circuit of wire induces an emf in that circuit. Actually, it is the rate of change in the number of field lines passing through a circuital loop that determines the induced emf in that process.*

The verbal statement of Faraday's law of induction is said to be that the induced emf in a circuit is equal to the rate at which the magnetic flux through the circuit is changing with time.

Mathematically the definition of Faraday's law is written by

$$\varepsilon = -\frac{d\Phi_B}{dt}, \tag{4.1}$$

where ε is the induced emf and Φ_B is the magnetic flux that passing through the circuit. The negative sign indicates the direction of induced emf and the physical significance of it is described the next section. If the coil consists of N turns, then an induced emf appears in every turn, and the total induced emf in the circuit is the sum of the individual values.

Hence, the total induced emf is given by

$$\varepsilon = -N\frac{d\Phi_B}{dt}. \tag{4.2}$$

Note: The magnetic flux Φ_B to be a measure of the number of magnetic force lines crossing through a surface area. The magnetic flux through a surface is defined by the similar formula of an electric flux stated in Eq. 1.6 and given by

$$\Phi_B = \int \vec{B}.d\vec{A}. \tag{4.3}$$

The SI units of induced emf and magnetic flux are *Volt* and *Weber*, respectively. $1\ Weber = 1\ Tesla.meter^2$.

4.3 Lenz's Law

Significance of negative sign of Faraday's law:

In Eq. 4.1 we see that a negative sign has been used which indicates the direction of induced emf. In this section we shall discuss the physical significance, that why a negative sign has been chosen in Faraday's law of induction. In 1834 Heinrich Lenz proposed a rule for the direction of induced current that produced in the coil and it is known as Lenz's law.

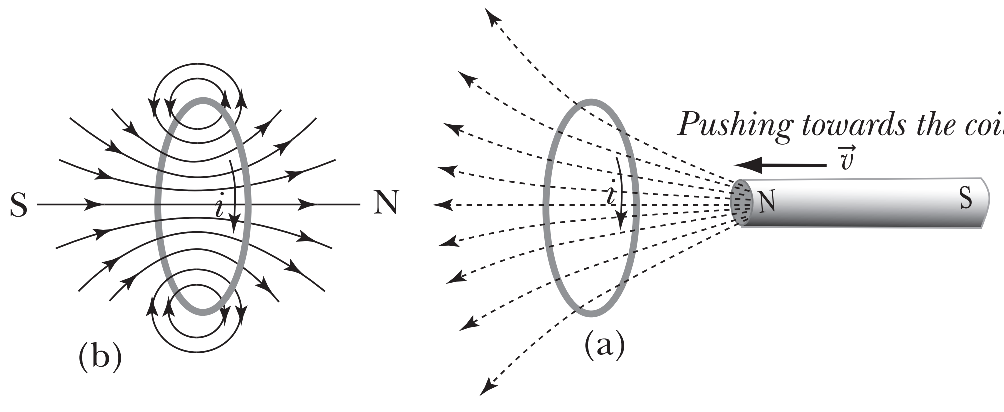


***Figure 4.3:** Indication of direction of induced current in the loop. (a) Due to pushing the magnet towards the loop, the magnetic flux through the loop is increasing. (b) Due to the change of flux through the loop, the induced current i is produced in such a direction that the magnetic field produced by induced current i opposes the pushing of the bar magnet.*

Figure 4.3(a) shows the N pole of a bar magnet is pushing towards a conducting loop. Due to pushing the magnet, it is increasing the magnetic flux with time. According to Faraday's law, the change of magnetic flux through the loop with time will induce an emf that creates a current in the loop. Practically, it is experienced that there is a appearance of force which opposes the pushing of the bar magnet toward the loop. According to Lenz's law this opposition force comes into appearance due to the direction of induced current i. The direction of i is shown by arrow markers in figures. Figure 4.3(b) shows that the induced current i sets up a magnetic field in the loop by the system of right-hand rule. It is clear that the direction of i appears in such that it creates an N pole facing towards the N pole of the bar magnet. By the process of repulsion of N-N poles, the creating situation finally opposes the pushing of the bar magnet towards the loop.

Again let us now imagine the Fig 4.4(a). It describes that the N pole of a bar magnet is pulling away from the conducting loop. Due to pulling away the magnet, it is decreasing the magnetic flux with time. Practically, the experienced force opposes the pulling away of the bar magnet from the loop. Lenz proposed regarding this opposition force comes into appearance due to the opposite direction of induced current i as it is stated earlier. Figure 4.4(b) shows that the induced current i sets up a magnetic field also by the system of right-hand rule. Here, the direction of i is such that it creates an S pole facing towards the N pole of the bar magnet. By the process of attraction of N-S poles, the creating situation finally opposes the pulling away of the bar magnet from the loop.

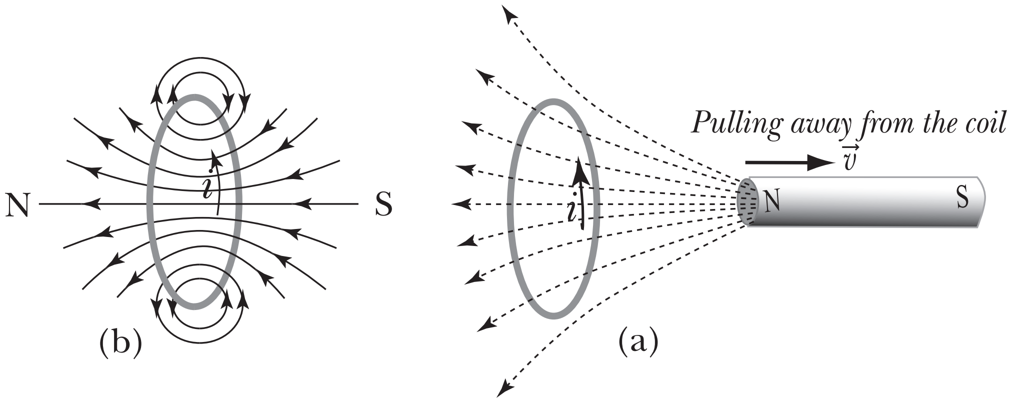


Figure 4.4: ***Indication of direction induced current in the loop. (a) Due to pulling the magnet away from the loop, the magnetic flux through the loop is decreasing. (b) Due to the change of flux through the loop, the induced current*** i ***is produced in such a direction that the magnetic field produced by induced current*** i ***opposes the pulling of the bar magnet.***

However, Fig 4.3 describe the direction of induced current in the loop which opposes the increasing of the change of flux and Fig. 4.4 describe the direction of induced current which opposes the decreasing of the change of flux through the loop. We see that in both cases *the flux of the magnetic field produced by the induced current opposes the change of flux that causes the induced current.* This unavoidable opposite factor of appearing induced current in the loop suggests the acceptance of negative sign in Faraday's law of induction.

4.4 Eddy Current

An induced current produces not only in a loop or coil, the induced effect is also observed if a metal sheet or plate is to be moved in a magnetic field. *When a metal sheet or plate is made to move in a magnetic field, due to the change of magnetic flux through the metal, an induced current appears in that metal sheet or plate. This induced current is called eddy current.* In most cases the eddy current is undesirable effect due to the lose of energy in the system. For example, they increase the internal energy and thus can increase the temperature of the material. For this reason, materials that are subject to changing magnetic field are often laminated or insulated. In some cases eddy-current heating effect can be used to advantage, as in an induction furnace, in which a sample of material can be heated using a rapidly changing magnetic field. Figure 4.5 shows when a conducting plate is taking away from the magnetic field, an eddy current is produced can be shown by a current loop.

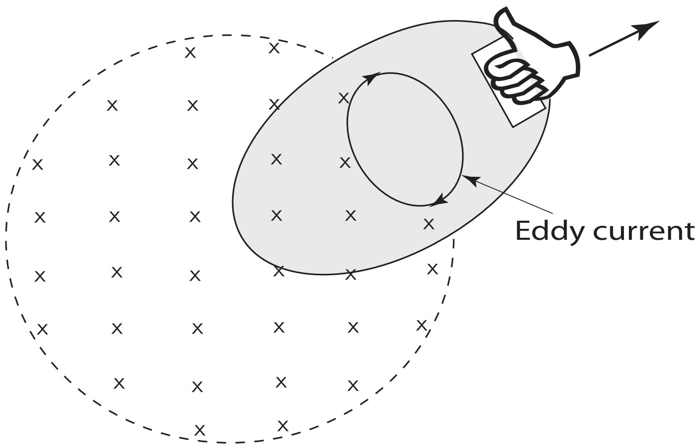


Figure 4.5: Eddy current is produced in a conducting material when it moves in magnetic field.

The effective eddy current path experience a magnetic force $\vec{F}_B = i\vec{L} \times \vec{B}$, where L is the effective length of the current path. This force is transmitted to the material and according to Lenz's law it opposes the motion of the conductor. This effect gives rise a deceleration in a rotating wheel or a moving track operated in magnetic field.

4.5 Inductance

An inductor is an ideal solenoid, carrying a current i that produces an magnetic field $\vec{B}$ in its interior position. An inductor stores energy in the form of magnetic field surrounding its current carrying wires, just like as a capacitor stores energy in the form of electric field between its charged plates. Inductance is the property of an inductor depends on its geometry, like the capacitance, the property of a capacitor also depends on its geometry. In an electrical circuit diagram, an inductor is represented by the symbol —ɑɑɑɑɑ—, which pictures the shape of a solenoid.

Mathematical representation:

Whenever we change the current in a solenoid, it changes the magnetic field. Faraday's law shows that there is an emf generated in the solenoid directly proportional to the rate of change of current.

$$\varepsilon_L \propto \frac{di}{dt},$$

$$\text{or} \qquad \varepsilon_L = L\frac{di}{dt}, \tag{4.4}$$

where L is a proportionality constant and called inductance. If the solenoid has N number of turns, it sets a flux $N\Phi_B$ and is known as the number of *flux linkages* of the inductor. According to Faraday's law, the emf is given by

$$\varepsilon_L = -\frac{d(N\Phi_B)}{dt}. \tag{4.5}$$

From the Eqs. 4.4 and 4.5 (taking the magnitude) we have

$$L\frac{di}{dt} = \frac{d(N\Phi_B)}{dt}.$$

Integrating respect to time

$$Li = N\Phi_B + C,$$

where C is the integrating constant. Assuming that when $i = 0$, then $\Phi_B = 0$ and $C = 0$. Therefore, we find

$$Li = N\Phi_B,$$

$$\text{or} \qquad L = \frac{N\Phi_B}{i}. \tag{4.6}$$

Note that, since Φ_B is proportional to the current i, the Eq. 4.6 indicates the inductance (like the capacitance) depends only on the geometry of the device.

4.5.1 Self inductance

Suppose that a steady current i, supplied by a battery, exists in a coil. Let us suddenly the current in the circuit is reduced to zero. This will cause the change of magnetic field and an induced emf will appear in such a direction, according to the Lenz's law it will oppose the decrease in current. Again if the current i in the circuit is to be increased, the induced emf will oppose to increase the current in that circuit. This effect in that circuit created itself is called self-inductance.

However, when we talk about inductance, it refers the self-inductance and is also denoted by L.

Unit of inductance:

Inductance is written as with SI unit

$$L = \frac{\varepsilon}{\frac{di}{dt}} \cdot \frac{\text{volt}}{\frac{\text{ampere}}{\text{second}}} = \frac{\varepsilon}{di/dt}\ \text{henry}.$$

$\therefore$ 1 henry= 1 volt.second/ampere.

4.5.2 Mutual inductance

If current i is set up in primary coil P from zero to a maximum value, the change of flux sets up an induced emf in the secondary coil S to maximum. This induced emf sets a current in S. Again if current in P is cut-off suddenly, then current in S will change, which causes the change in flux linkage in P and sets up an induced emf in P. In this case the induced emf in P sets up a current in opposite direction of the former current in S. The presence of the secondary coil in the neighborhood of the primary coil makes a delay of the growth of current. Again when the current is made to decrease in P, due to the presence of secondary coil S, it makes also delay to decrease the current in P. This mutual action of this two coil is called mutual induction between them.

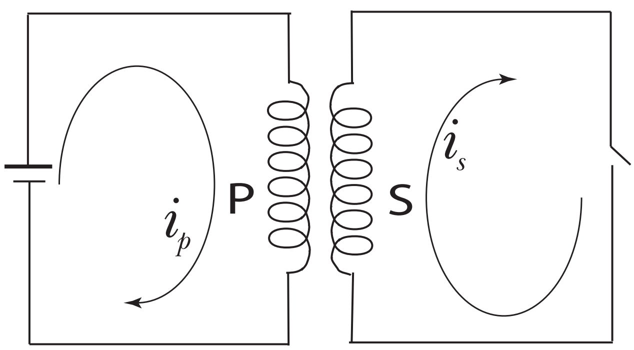


Figure 4.6: Mutual inductance creates between two coils situated close together. P and S are called primary and secondary coils, respectively.

Let a current i_p is sets up in P, which creates a magnetic field B_p in it. This $\vec{B}_p$ produces a flux in S. Therefore, the flux in secondary coil S due to primary coil P can be calculated by

$$\Phi_{sp} = \int \vec{B}_p.d\vec{A}_s,$$

where A_s is the area of single turn of S. If there is N_s turns of S, then

$$N_s\Phi_{sp} \propto i_p,$$

$$\text{or} \qquad N_s\Phi_{sp} = M_{sp}i_p,$$

where M_{sp} is the mutual inductance induces in S coil due to the P coil.

This flux produces an emf in S due to P, is given by

$$\varepsilon_{sp} = -N_s\frac{d\Phi_B}{dt} = -M_{sp}\frac{di_p}{dt} \tag{4.7}$$

This induced emf ε_{sp} in secondary coil S, causes a flow of current i_s in that coil, and similarly produces a mutual inductance M_{ps} in P.

Therefore, the induced emf in P due to coil S is to be calculated as

$$\varepsilon_{ps} = -M_{ps}\frac{di_s}{dt} \tag{4.8}$$

For any two circuits close together, in fact

$$M_{sp} = M_{ps} = M,$$

where M is generally called the mutual inductance. *The unit of mutual inductance is also denoted by henry.*

4.5.3 The inductance of a solenoid

Consider a cross section of a solenoid of length l, cross sectional area A, number of turns per unit length be n and carrying a current i. We have seen, in Subsection 3.7.2, the magnetic field B inside a solenoid carrying a current i has been shown

$$B = \mu_0 ni.$$

The number of flux linkage in the length l is

$$N\Phi_B = (nl)(BA).$$

Now substituting the value of B, we have

$$N\Phi_B = \mu_0 n^2 liA \tag{4.9}$$

Now using Eq. 4.6, the inductance for a solenoid can be calculated

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 n^2 liA}{i} = \mu_0 n^2 lA, \tag{4.10}$$

and the inductance per unit length of the solenoid can written by

$$\frac{L}{l} = \mu_0 n^2 A. \tag{4.11}$$

Equation 4.10 shows that the inductance of a solenoid does not depends on the current or the magnetic field. It depends on geometrical factors number of turns n, length l and cross sectional area A.

4.5.4 The inductance of a toroid

Figure 4.7 shows a rectangular cross section of a toroid. The toroid has outer radius 'b', inner radius 'a' and height of the rectangular cross section 'h'. The magnetic field in a toroid is given by Eq. 3.27

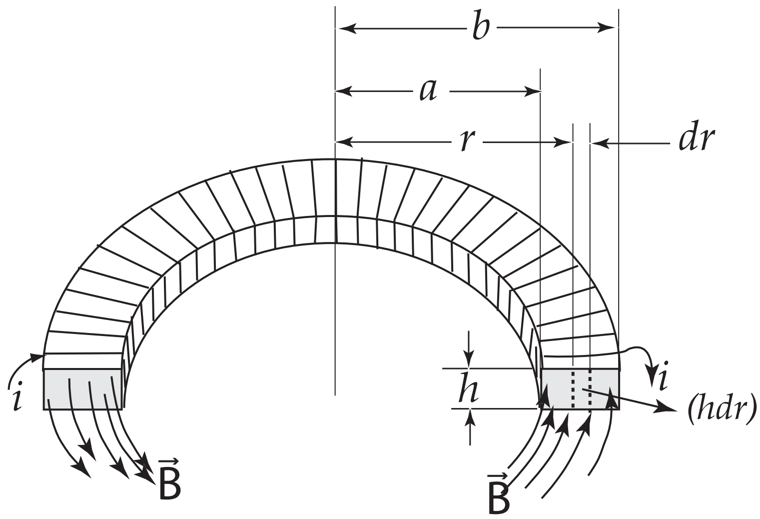


Figure 4.7: A cross section of a toroid. $\vec{B}$ concentric magnetic field, 'a' and 'b' are inner and outer radius respectively. The thickness of the toroid is 'h'.

$$B = \frac{\mu_0 i N}{2\pi r},$$

where we see that the magnetic field inside the toroid is not constant and varies with the radius r.

Let us take a small cross sectional area of the toroid be hdr. The total flux Φ_B through the cross section is

$$\Phi_B = \int \vec{B}.d\vec{A} = \int_a^b B(hdr) = \int_a^b \frac{\mu_0 i N}{2\pi r} hdr$$

$$= \frac{\mu_0 i N h}{2\pi} \int_a^b \frac{dr}{r} = \frac{\mu_0 i N h}{2\pi} ln\frac{b}{a}.$$

The inductance of the toroid can be found by the equation

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 h}{2\pi} ln\frac{b}{a}. \tag{4.12}$$

Here we also see the inductance L depends only on geometrical factors.

4.6 Examples

Example 4.1 The long solenoid S of Fig. 4.8 has 220 turns/cm; its diameter d is 3.2 cm. At its center we place a 130-turn close-packed coil C of diameter d_c=2.1 cm. The current in the solenoid is increased from zero to 1.5 A at a steady rate over a period of 0.16 s. What is the magnitude of the induced emf that appears in the central coil while the solenoid is being changed?

Solution: Since the current is changing with time, the magnetic field produced by the current is also changing in the interior points of the solenoid. The field is uniform and perpendicular to the area of each turn of coil C.

$$\Phi_B = \int \vec{B}.d\vec{A} = BA,$$

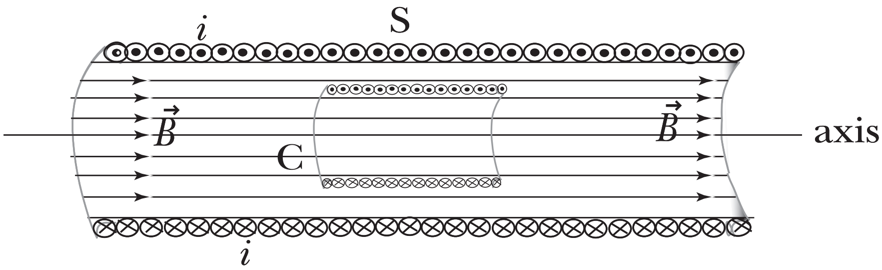


***Figure 4.8:** A coil 'C' is placed inside a solenoid 'S'. The solenoid carries a current that emerges from the page at the top and enters at the bottom, as indicated by the dots and crosses. When the current is changing in the solenoid, an induced emf appears in the coil.*

where A is the area of each turn of the coil C.

Now from Faraday's law of induction followed by Eq. 4.2 with absolute value

$$|\varepsilon| = N\frac{d\Phi_B}{dt} = N\frac{d(BA)}{dt} = NA\frac{d(B)}{dt} = NA\frac{d(\mu_0 ni)}{dt} = NA\mu_0 i\frac{di}{dt}.$$

Area of the single turn of the central coil,

$$A = \frac{\pi d_c^2}{4} = \frac{\pi(0.021)^2}{4}m^2 = 3.4636\times 10^{-4}m^2.$$

$N = 130$, $\mu_0 = 4\pi\times 10^{-7}$ T.m/A, $n = 2.2\times 10^4$ turns/m. $di = (1.5-0) = 1.5$ A and $dt = 0.16$ s.
Therefore,
$|\varepsilon| = (130)(3.4636\times 10^{-4}$ m^2) ($4\pi\times 10^{-7}$ T.m/A)(2.2×10^4 turns/m) (1.5 A/0.16 s)
$= 1.2\times 10^{-2}$ V =12 mV.

Example 4.2 A long solenoid of diameter 5 cm has 100 turns/cm of its length. At the center of the solenoid, a small coil of 50 turns and 2 cm diameter is placed such that the magnetic field produced by the solenoid is parallel to the axis of the coil. The current is changed from -2.0 A to +2.0 A over a period of 0.1 second. Calculate the emf induced in the coil during the change.

Solution: Faraday's law of induction

$$|\varepsilon| = N\frac{d\Phi_B}{dt} = N\frac{d(BA)}{dt} = NA\frac{d(B)}{dt} = NA\frac{d(\mu_0 ni)}{dt} = NA\mu_0 i\frac{di}{dt}.$$

Area of the single turn of the central coil,

$$A = \frac{\pi d_c^2}{4} = \frac{\pi(0.02)^2}{4}m^2 = 3.1416\times 10^{-4}m^2.$$

$N = 50$, $\mu_0 = 4\pi\times 10^{-7}$ T.m/A, $n = 1\times 10^4$ turns/m.
$di = \{2-(-2)\}$ A=4 A and $dt = 0.1$ s.
Therefore,
$|\varepsilon| = (50)(3.1416\times 10^{-4}$ m^2) ($4\pi\times 10^{-7}$ T.m/A)(1×10^4 turns/m) (4 A/0.1 s)
$= 7.88\times 10^{-3}$ V =7.88 mV.

Example 4.3 A section of a solenoid of length $l = 12$ cm and having a circular cross section of diameter $d = 1.6$ cm carries a steady current of $i = 3.80$ A.The section contains 75 turns along the length. (a) What is the inductance of the solenoid when the core is empty? (b) the current is reduced at a constant rate to 3.20 a in a time of 15 s. What is the resulting emf developed by the solenoid?

Solution: (a) Form Eq. 4.10, we have the inductance of a solenoid is

$$L = \mu_0 n^2 lA.$$

Here, n=75 turns/0.12 m and A=$\pi(0.008)^2$.

Therefore,

$$L = (4\pi \times 10^{-7} H/m)(75turns/0.12m)^2(0.12m)\pi(0.008m)^2 = 1.184 \times 10^{-5} H = 11.84\mu H.$$

(b) The rate at which the current changes is

$$\frac{di}{dt} = \frac{3.20A - 3.80A}{15s} = -0.04A/s.$$

The induced emf in magnitude

$$\varepsilon = \left| L\frac{di}{dt} \right| = (11.84\mu H)(0.04A/s) = 0.4737\ \mu V.$$

Example 4.4 A toroid having square cross section with a side 5 cm. It contains 536 turns, inner radius 15.3 cm and carries a current 810 mA. Calculate magnetic flux, total flux-linkage and inductance.

Solution: We have inner radius a=15.3 cm, h=50 cm and outer radius b=(15.3+5) cm =20.3 cm. The magnetic flux

$$\Phi_B = \frac{\mu_0 iNh}{2\pi} ln\frac{b}{a}$$

$$= \frac{(4\pi \times 10^{-7} H/m)(0.810A)(536turns)(0.05m)}{2\pi} ln\frac{0.203m}{0.153m} = 1.2277 \times 10^{-6} weber.$$

Total flux linkage

$$N\Phi_B = \frac{\mu_0 iN^2h}{2\pi} ln\frac{b}{a} = 6.5803 \times 10^{-4} weber.$$

Finally the inductance of the toroid is to be

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 h}{2\pi} ln\frac{b}{a} = 8.1238 \times 10^{-4} H = 0.81238\ mH.$$

Example 4.5: A circular loop made of a stretched conducting elastic material has a 1.23 m radius. It is placed with its plane at right angles to a uniform 785 mT magnetic field. When released, the radius of the loop starts to decrease at an instantaneous rate of 7.50 cm/s. Calculate the emf induced in the loop at that instant.

Solution: The magnetic field is perpendicular to the surface area. So, $\vec{B}.d\vec{A} = B\ dA\ \cos 0^o = B\ dA$.

$$\therefore \quad \Phi_B = \int \vec{B}.d\vec{A} = \int B\ dA = BA = B(\pi r^2),$$

where r is the radius of the loop. When the loop is released then the radius starts to decrease, hence area A starts to decrease which produce the change of magnetic flux.

$$\therefore \quad d\Phi_B = B\ dA = B\ d(\pi r^2) = 2\pi r B\ dr.$$

Now the induced emf is

$$\varepsilon = -\frac{d\Phi_B}{dt} = -2\pi r B\frac{dr}{dt}.$$

It is given that $B = 0.785\ T$, $r = 1.23\ m$, and $\frac{dr}{dt} = -7.50 \times 10^{-2}\ m/s$ (here negative sign indicates the decreasing of the radius).

$$\text{Then} \qquad \varepsilon = -2\pi r B\frac{dr}{dt} = 2\pi(1.23\ m)(0.785\ T)(-7.50 \times 10^{-2}\ m/s)$$

$$= 0.455\ V.$$

Chapter 5

MAGNETIC PROPERTIES OF MATER

Magnetic materials play important roles in our daily lives. Materials such as iron, which can be permanent magnets at ordinary temperatures, are commonly used in electric motors and generators as well as loudspeakers. Some materials are analogous to dielectric in that they acquire an induced magnetic field in response to an external magnetic field and the induced field vanishes when the external field is removed. The materials which can be magnetized and demagnetized easily are widely used for storing information in such applications as magnetic recording tape (audio and video tape recorders), computer disks, bank and credit cards etc.

In the starting of this chapter, we shall introduce the internal structure of materials that is responsible for their magnetic properties. We shall show that the behavior of different magnetic materials can be understood in terms of the magnetic dipole moments of individual atoms. A vast understanding of magnetic properties of mater requires quantum mechanical discussions. But in this chapter we shall not enter to the interior of quantum mechanics. We wish to discuss qualitative understanding of magnetic properties of mater in the chapter.

5.1 The Magnetic Dipole

In the Sec. 1.2 of Chapter 1, we have seen the configuration of two static charges as *electric dipole* and how it can produce an electric field (Fig. 1.4). For static electric fields, the single isolated charge is the fundamental quantity. Individual charges produce an electric field, and in turn the electric field set up by one group of charges can influence the behavior of other charges. The common phenomena regarding this are the interaction between nucleus and electrons in an atom, the force exserted by one atom on another atom in ionic molecules and solids etc.

For steady magnetic field, the fundamental quantity is the moving electric charges in a current element, which can produce produce a magnetic field and can also influenced by other magnetic field. However, in trying to magnetic properties of materials that how moving charges in the atoms can produce magnetic dipoles. Actually the magnetic properties of materials to be the collection of atoms with individual magnetic dipole moments.

Let us begin by considering the magnetic field due to a circular loop (Subsection 3.4.2 Eq. 3.15) at a point on the axis ‘x’ is

$$B = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}.$$

We have far from the loop (if, $x >> R$)

$$B = \frac{\mu_0 i R^2}{2x^3},$$

$$\text{or} \qquad B = \frac{\mu_0}{2\pi}\frac{i\pi R^2}{x^3}. \tag{5.1}$$

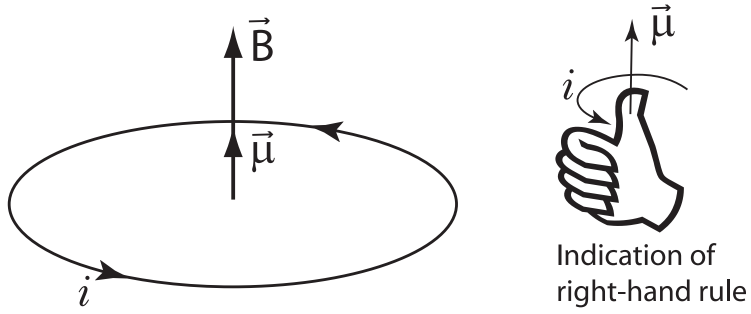


***Figure 5.1:** The magnetic dipole moment $\vec{\mu}$ and field vector $\vec{B}$ along the axis which is perpendicular to the plane of the current loop.*

The quantity $i\pi R^2$ in Eq. 5.1 can be written as iA, where $A = \pi R^2$ is the area of the circular loop. We define this quantity to be the magnitude of the *magnetic dipole moment* μ of the loop:

$$\mu = iA. \tag{5.2}$$

The magnetic dipole moment of a current loop is the product of the current and the area of the loop. If the loop has N turns, the magnetic dipole moment be

$$\mu = NiA. \tag{5.3}$$

Like the electric dipole moment, the magnetic dipole moment is a vector quantity. The direction of $\vec{\mu}$ is perpendicular to the plane of the current loop, determined by the right-hand rule as it is shown in Fig. 5.1. Therefore, Eq. 5.1 can be rewritten in vector notation as

$$\vec{B} = \frac{\mu_0 \vec{\mu}}{2\pi x^3}. \tag{5.4}$$

Note that $\vec{B}$ and $\vec{\mu}$ are vectors in the same direction and it is also shown in Fig. 5.1.

Let us now see the effect of a magnetic dipole placed in a magnetic field. Figure 5.2 shows a current loop placed in a uniform magnetic field $\vec{B}$ produced by some external agent. We know that

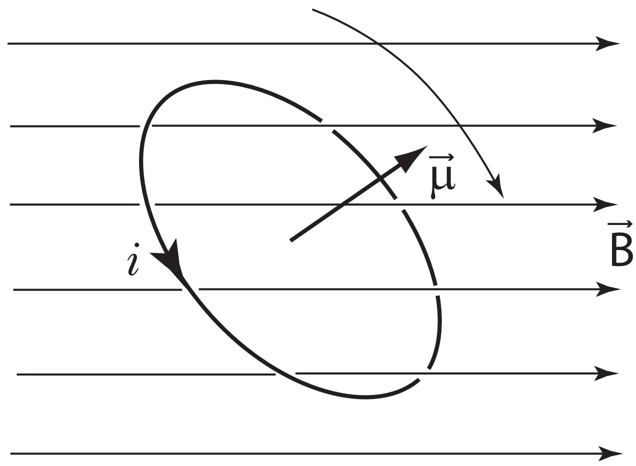


***Figure 5.2:** A magnetic dipole moment experiences a torque that rotates $\vec{\mu}$ into alignment with $\vec{B}$. $\vec{B}$ is applied by an external agent.*

a current loop in a uniform magnetic field experiences no net force but does experience a net torque given by

$$\vec{\tau} = iA\hat{n} \times \vec{B},$$

where $\hat{n}$ is a unit vector perpendicular to the loop. Since $\hat{n}$ and $\vec{\mu}$ have the same direction, we can write

$$\vec{\mu} = iA\hat{n}.$$

Therefore, the current loop experiences a net torque and can be expressed by the formula

$$\vec{\tau} = \vec{\mu} \times \vec{B}. \tag{5.5}$$

This torque tends to rotate the loop so that $\vec{\mu}$ lines up with the magnetic field vector $\vec{B}$. Equation 5.5 has the similar characteristics for the torque that an electric dipole in an electric field as given $\vec{\tau} = \vec{p} \times \vec{E}$.

Equation 5.4 and 5.5 give us two basic idea of magnetic properties of matter: Eq. 5.4 indicates how a magnetic field is produced by a magnetic dipole, and Eq. 5.5 shows how a magnetic dipole is influenced by an applied magnetic field. These two concepts will help us to understand the magnetic behavior of materials.

[If the magnetic dipole moment $\vec{\mu}$ makes an angle θ with $\vec{B}$, then we can write the potential energy as

$$U = -\vec{\mu}.\vec{B} = -\mu B \cos\theta. \tag{5.6}$$

The result is similar for an electric dipole moment, energy $U = -\vec{p}.\vec{E}$. In Eq. 5.6 $U = 0$ when $\theta = 90^o$ ($\vec{\mu}$ is perpendicular to $\vec{B}$). U has its smallest value ($= -\mu B$) when $\vec{\mu}$ and $\vec{B}$ are parallel, and has largest value ($= \mu B$) when $\vec{\mu}$ and $\vec{B}$ are antiparallel.]

Many physical system have magnetic dipole moments: the earth, bar magnets, current loops, atoms, nuclei, and elementary particles. Nuclear magnetic dipole moments are typically three to six order smaller in magnitude than atomic dipole moment. Ordinary magnetic effects in materials are determined by atomic magnetism. To exert a particular torque necessary to align nuclear dipoles requires a magnetic field about three to six orders larger that that necessary to align atomic dipoles.

5.1.1 The field of a dipole

Here we show the magnetic field pattern of magnetic dipole of a current loop and compare the analogy with an electric dipole and a dipole of a bar magnet. We have shown the complete field pattern of an electric dipole with the Fig. 1.4 discussed on page 3. A few lines for an electric dipole are shown in Fig. 5.3(a) and can be compared with the field lines for a current loop shown in Fig. 5.3(b). We see the great similarity between the patterns of field lines outside the loop. But we see a significant difference between electric and magnetic field lines is that electric field lines start on positive charges and end on negative charges, whereas magnetic field lines start on the plane of the closed loop and ends on the same plane. Actually the magnetic field lines always form closed loops.

Figure 5.3(c) shows the field lines of a bar magnet. It shows the same pattern of the field lines as the current loop, so a bar magnet can also be considered to be a magnetic dipole. From the outside view of the bar magnet, it seems that the force lines are leaving the N pole and converging on the S pole. The assuming N and S poles are to behave like the positive and negative charges of an electric dipole. But the actual situation is different instead of N and S poles concept of a bar magnet. The interior inspection of Fig. 5.3(c) shows that the field lines do not start and end on the poles but continue through the interior of the magnet, again forming closed loops. Therefore, N and S poles do not behave like the charges in an electric dipole and there is no real existence of isolated N and S poles in a bar magnet. We know that one of the Maxwell's equation $\vec{\nabla}.\vec{B} = 0$ established by the conception as there is no isolated magnetic pole can be found in nature.

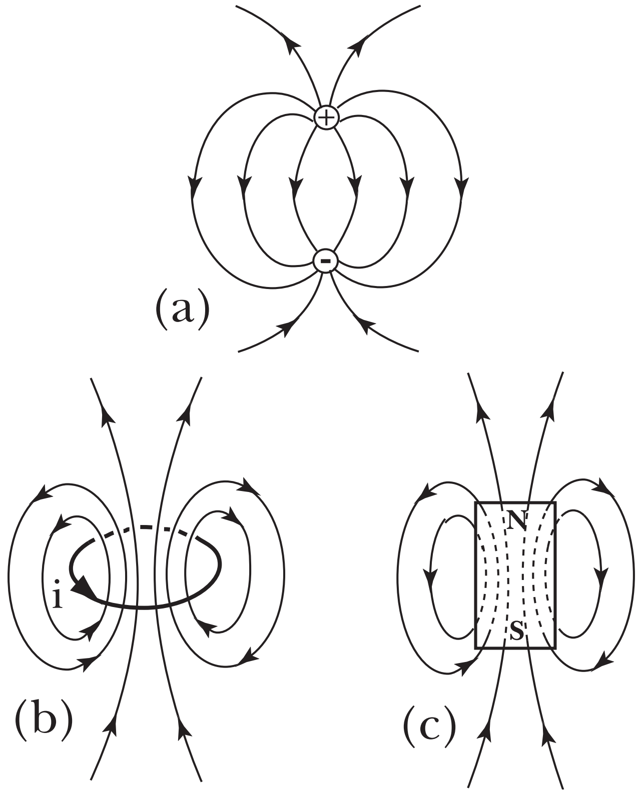


Figure 5.3: (a) Electric field configuration of an electric dipole. (b) The dipole magnetic field of a current loop. (c) The dipole magnetic field of a bar magnet.

5.1.2 Microscopic view of atomic magnetism

The magnetic properties of materials depend on the magnetic dipole moments of individual atoms, and we can consider magnetic materials to be composed of a collection of atomic dipoles, which might align when an external magnetic field is applied, as it is shown in Fig. 5.4. In practise we

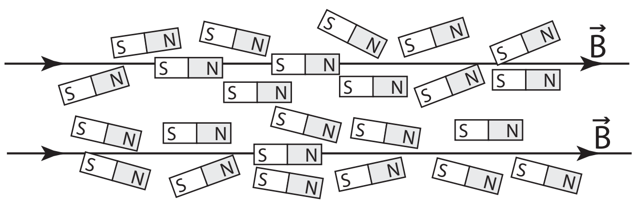


Figure 5.4: A magnetic materials can be considered as a collection of magnetic dipole moments. Each dipole moment consists by a schematic north and a south pole. In microscopic view, each dipole is actually a current loop that can not produce individual magnetic poles.

can not break apart the atoms into separate N and S magnetic poles, like the individual charges of an electric dipole.

We consider the magnetic moments to be tiny current loops, caused for example by the circulation of electrons in orbits in the atom. Figure 5.5(a) shows the current loop and magnetic field pattern of an atomic magnetism. Figure 5.5(b) shows the schematic drawing of an atomic magnet which represents the analogy of Fig. 5.5(a). Now we are going to discuss the magnetic dipole moment associated with a circulating electron.

We consider a simple model of an atom in which an electron moves in a circular orbit of radius r and speed v about the nucleus. The circulating electron can be considered as a current loop. If T

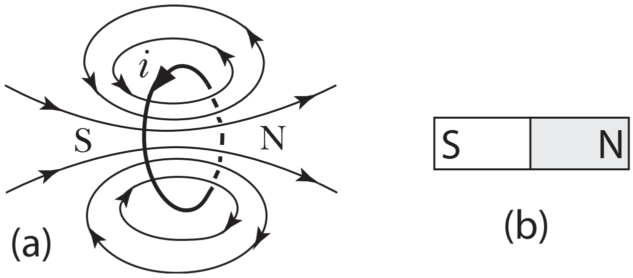


***Figure 5.5:** (a) Magnetic dipole shown by a tiny current loop, the current loop is formed by the circulation of orbital electrons in the atom. (b) Schematic drawing of magnetic dipole of an atom constitutes with N and S poles.*

be the revolving time of the electron then current

$$i = \frac{e}{T} = \frac{e}{2\pi r/v}. \tag{5.7}$$

Now the magnetic dipole moment can be found

$$\mu = iA = \left(\frac{ev}{2\pi r}\right)(\pi r^2) = \frac{evr}{2}. \tag{5.8}$$

Here the magnetic dipole moment is known as the *orbital magnetic dipole moment.* Because it is due to the orbital motion of electrons about the nucleus. The Eq. 5.8 can be rewritten as

$$\mu_l = \frac{evr}{2} = \frac{e}{2m}(mvr) = \frac{e}{2m}\,l, \tag{5.9}$$

where m is the mass of electron, l is the angular momentum of the electron moving in a circular orbit about the nucleus of the atom and μ_l dipole moment due to angular momentum l. According to the theory of quantum mechanics, the angular momentum is quantized and defined as $l = h/2\pi$, where h is the Planck constant. Substituting this fundamental value of angular momentum into Eq. 5.9, we obtain a basic unit of the dipole magnetic moment called the *Bohr magnetron* μ_B:

$$\mu_B = \frac{e}{2m}\frac{h}{2\pi} = \frac{eh}{4\pi m} = 9.27\times10^{-24}\ J/T, \tag{5.10}$$

where the numerical value is obtained by substituting the numerical value of e, h and m into Eq. 5.10. Atomic magnetic moments are usually measured in Bohr magnetron and are typically on the order of $1\mu_B$ in magnitude.

Besides the orbital dipole moment, electrons themselves have dipole moments called the *intrinsic or spin* magnetic moment. Electrons in different states of motion have diffract orbital magnetic dipole moments, but all electrons have exactly the same intrinsic magnetic dipole moment. The intrinsic magnetic moment of an electron is near equal to 1 μ_B.

The orbital and spin magnetic dipole moments of electrons are about the same size (of 1 μ_B), and thus both are important in determining the magnetic properties of atoms. The total magnetic moment of an atom is obtained from the vector sum of the orbital and spin magnetic moments of all its electrons. In an atom with many electrons, these vector sums may be very complicated. In some atoms, however, the total orbital and spin magnetic moments turn to be zero. Materials made from these atoms are virtually nonmagnetic and show only a weak induced effect called *diamagnetism.* In other atoms, the total orbital or spin or both magnetic moment may be nonzero, so that atoms align in the presence of an external magnetic field. These materials are called *paramagnetism.* The most familiar type of magnetic behavior is *ferromagnetism.* The total magnetic moments of ferromagnetic materials are nonzero but have a significant magnitude. The alignment of the atoms remains even when the external field is removed.

5.2 Magnetization

When a substance is placed in an external magnetic field, all atoms of the substance may acquire an induced magnetic dipole moment due to alignment of the individual dipoles. The process of alignment of the dipoles to acquire an induced magnetic dipole moment is known as magnetization. Consider a specimen of volume V consisting n atoms having magnetic dipole moments $\vec{\mu}_n$. The net dipole moment $\vec{\mu}$ can be computed as

$$\vec{\mu} = \sum \vec{\mu}_n.$$

We then define the magnetization $\vec{M}$ in vector from to be the net dipole moment per unit volume,

$$\vec{M} = \frac{\vec{\mu}}{V} = \frac{\sum \vec{\mu}_n}{V}. \tag{5.11}$$

If the magnetization to be uniform then the substance has a continuous dipole moment. In this condition the magnetization to be considered a microscopic quantity and written as

$$\vec{M} = \lim_{\Delta V \to 0} \frac{\sum \vec{\mu}_n}{V}. \tag{5.12}$$

5.2.1 Diamagnetism

In 1847, Michael Faraday discovered that a specimen of bismuth was repelled by a strong magnet. (Whereas paramagnetic substances are always attracted by a magnet.) Diamagnetic substances becomes weakly magnetized when placed in a magnetic field and resultant magnetization is opposed to the applied field. The atomic dipole moments of such materials are zero. Perhaps atoms having several electrons with their orbital and spin magnetic moments adding vectorially to zero. Diamagnetic substances have no permanent dipole moments and only acquire dipole moments when it placed in an external magnetic field. Examples of diamagnetic materials are: bismuth, Zn, Ag, Cu, Sb, Pb, Au.

5.2.2 Paramagnetism

Paramagnetic substance are those which, when placed in a magnetic field, become weakly magnetized in the direction of the applied field. Paramagnetic substances have permanent atomic dipole moments and initially are randomly oriented in space. Paramagnetic materials are weakly attracted by the external magnetic field. When the external magnetic field is removed from a paramagnetic sample, the thermal motion causes the directions of the magnetic dipole moments to become random again. So that the paramagnetic materials do not become permanent magnets. Examples of paramagnetic materials are: Mn, Pt (Platinum), Pd (Palladium) etc.

5.2.3 Ferromagnetism

Ferromagnetic substances are those which, when placed in a magnetic field, becomes strongly magnetized in the direction of applied field. The atoms of ferromagnetic materials have permanent dipole moments. Ferromagnetic materials sometimes classed as strongly paramagnetic. All ferromagnetic materials are paramagnetic but all paramagnetic substances are not ferromagnetic. When an external field is applied in ferromagnetic materials, a strong interaction occur that the materials keep their atomic dipole moments aligned even when the external field is removed. The ferromagnetism disappears if the the temperature is increased above a certain critical value called *Curie temperature* and substances become paramagnetic. The temperature at which a ferromagnetic material becomes paramagnetic is called its Curie temperatre. Familiar ferromagnetic materials at room temperature are iron, cobalt and nickel The Curie temperature of iron is 700^oC; above this temperature iron becomes paramagnetic

5.3 Three Magnetic Vectors

When a ferromagnetic material is placed in an external magnetic field, it to be strongly magnetized. This external field (say $\vec{H}$) magnetizes the magnetic material and makes alignment the atomic dipole moments to the direction of applied field. Let the magnetization vector of the ferromagnetic materials be $\vec{M}$. The two vectors, one is applied field vector $\vec{H}$ and another is $\vec{M}$, interact to each other in the material and gives a resultant magnetic field vector, can be denoted by $\vec{B}$. We now try to make a relation of these three magnetic vectors.

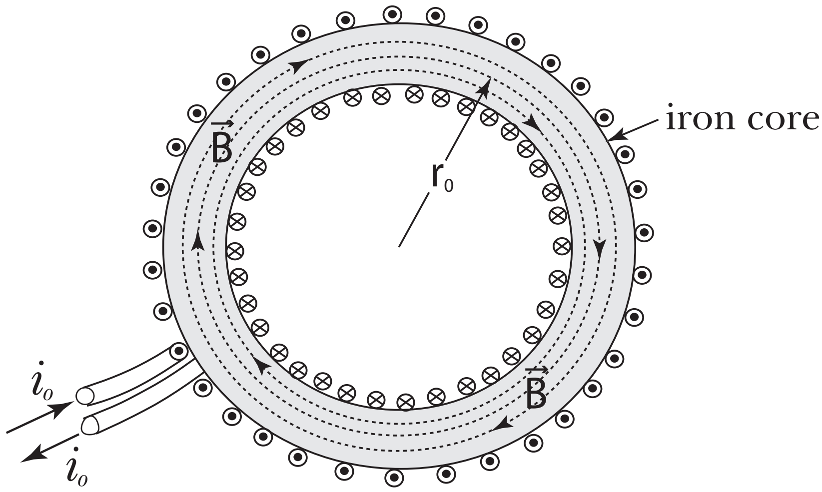


Figure 5.6: ***A toroid with core of ferromagnetic (iron) material. Current in each winding is*** i_0.

Let us consider a toroidal coil carrying a current i_o in it. Let a small length of the toroid be $d\vec{l}$ at a radial distance r_o. First consider the toroid without iron core, and according to Ampere's law we have

$$\oint \vec{B}.d\vec{l} = \mu_o i. \tag{5.13}$$

If the total turns of the toroid be N_o, then the Eq. 5.13 can be written as

$$B\,(2\pi r_o) = \mu_o i_o N_o. \tag{5.14}$$

Let us consider the existence of iron core. The value of $\vec{B}$ determined experimentally is different when an iron core is placed than without it. The net current in the toroid must be changed due to in placing the iron core. An additional amount of magnetized current i_m will be introduced in the system.

In the presence of magnetic material, the Ampere's law is modified by

$$\oint \vec{B}.d\vec{l} = \mu_o (i + i_m) \tag{5.15}$$

$$\text{or} \qquad B(2\pi r_o) = \mu_o i_o N_o + \mu_o i_{mo} N_o. \tag{5.16}$$

In fact i_{mo} is related to the magnetization $\vec{M}$. The current i_{mo} is the magnetization current due to contributions of individual atomic dipoles of the material.

Let the rectangular cross sectional area of the toroid be A. Then the volume of the element dl be Adl. The amount of dipole moment due to element dl can be found by the help of Eq. 5.11

$$d\mu = M\,(Adl). \tag{5.17}$$

But magnetic moment in a current loop is (see Eq. 5.3)

$$\mu = NiA. \tag{5.18}$$

For element dl the number of turns stands

$$\frac{N_o}{2\pi r_o}\,dl.$$

So considering the Eq. 5.18 we get another relation of dipole moment due to magnetization for element dl

$$d\mu = \frac{N_o}{2\pi r_o}\,dl.i_{mo}\,A. \tag{5.19}$$

Now comparing the Eqs. 5.17 & 5.19 we have

$$M\,(Adl) = \frac{N_o}{2\pi r_o}\,dl.i_{mo}\,A,$$

$$\text{or} \qquad N_o i_{mo} = M\,(2\pi r_o). \tag{5.20}$$

Substituting the value of $N_o I_{mo}$ in Eq. 5.16, one can get

$$B\,(2\pi r_o) = \mu_o i_o\,N_o + \mu_o\,M\,(2\pi r_o).$$

or in general

$$\oint \vec{B}.d\vec{l} = \mu_o i + \mu_o \oint \vec{M}.d\vec{l},$$

$$\text{or} \qquad \oint \left(\frac{\vec{B}-\mu_o\vec{M}}{\mu_o}\right).d\vec{l} = i. \tag{5.21}$$

Now we try to give the special name of the quantity $\frac{\vec{B}-\mu_o\vec{M}}{\mu_o}$, which is responsible for the magnetization of the material. We define it as the applied field intensity and denoted by H. Sometimes H is called the *magnetizing force* or *magnetizing intensity*. So, $\vec{H}$ can be written as

$$\vec{H} = \frac{\vec{B}-\mu_o\vec{M}}{\mu_o}, \tag{5.22}$$

Which follows

$$\vec{B} = \mu_o\vec{H} + \mu_o\vec{M}, \tag{5.23}$$

$$\text{or} \qquad \vec{B} = \mu_o\left(1+\frac{\vec{M}}{\vec{H}}\right)\vec{H}. \tag{5.24}$$

Equations 5.23 or 5.24 gives the relation between three magnetic vectors in a material placed in a magnetic field. $\vec{B}$ is the net magnetic field vector is equal to the sum of the applied field and magnetization vectors multiplied by μ_o. Now from Eqs. 5.21 and 5.22, we find Ampere's law in presence of magnetic material as

$$\oint \vec{H}.d\vec{l} = i, \tag{5.25}$$

where i is the current applied to the circuit and does not include the magnetizing current. The line integral of field intensity $\vec{H}$ around any closed path equals the total conduction current in the presence of magnetized material. Since in Eq. 5.25 i indicates the current applied by an external source, hence the field $\vec{H}$ is said to be magnetizing field.

5.4 Permeability & Susceptibility

Relative permeability:

For non-ferromagnetic materials, it is found experimentally that $\vec{B}$ is directly proportional to $\vec{H}$. That is, the proportionality is applicable only for diamagnetic and paramagnetic materials. Therefore,

$$\vec{B} \propto \vec{H},$$

$$\text{or} \qquad \vec{B} = \mu_o \kappa_r \vec{H}, \tag{5.26}$$

where κ_r is called relative permeability and μ_o is called simply permeability in vacuum. Vectors $\vec{B}$ and $\vec{H}$ have the same direction. Putting the value $\vec{B}$ of Eq. 5.26 in Eq. 5.23 we get

$$\mu_o \kappa_r \vec{H} = \mu_o \vec{H} + \mu_o \vec{M},$$

$$\text{or} \qquad \vec{M} = (\kappa_r - 1)\vec{H}. \tag{5.27}$$

Now in vacuum, the magnetization vector $\vec{M}$ is zero, because there is no magnetic dipole moment present. Hence putting $\vec{M} = 0$ in Eq. 5.23 we have

$$\vec{B} = \mu_o \vec{H}, \tag{5.28}$$

and in Eq. 5.27 it is found

$$(\kappa_r - 1) = 0,$$

$$\text{or} \qquad \kappa_r = 1.$$

Therefore, we see in vacuum the relative permeability κ_r is unity. For paramagnetic materials κ_r is slightly greater than unity and for diamagnetic materials κ_r is slightly less than unity.

Magnetic Susceptibility:

For para and diamagnetic substances the magnetization is proportional to the field intensity $\vec{H}$,

$$\text{or} \qquad \vec{M} \propto \vec{H},$$

$$\text{or} \qquad \vec{M} = \chi_m \vec{H}, \tag{5.29}$$

where χ_m is a proportionality constant and known as magnetic susceptibility. *Since $\chi_m = M/H$, it may be defined as the ratio of magnetization to the magnetizing field.* Comparing Eqs. 5.27 and 5.29 we get

$$\chi_m = \kappa_r - 1,$$

$$\text{or} \qquad \kappa_r = 1 + \chi_m. \tag{5.30}$$

Permeability:

Substituting κ_r of Eq. 5.30 in Eq. 5.26, we have

$$\vec{B} = (1 + \chi_m)\mu_o \vec{H},$$

or simply

$$\vec{B} = \kappa_m \vec{H}, \tag{5.31}$$

where $\kappa_m = \mu_o(1 + \chi_m)$ is the magnetic permeability of the medium. We see

$$\kappa_m = \mu_o(1 + \chi_m),$$

$$\text{or} \qquad \frac{\kappa_m}{\mu_o} = (1+\chi_m) = \kappa_r.$$

Therefore, the permeability of the medium is the measure of the total capacity of the medium to take up magnetization and the relative permeability κ_r of the medium is the measure of the ratio of permeability in the presence of materials to the absence of materials (free space). In other words, the relative permeability of the medium is defined as the ratio of the number of field per square meter crossing perpendicularly to that in air or free space.

5.5 Magnetic Hysteresis

We know that for diamagnetic and paramagnetic materials $\vec{B}$ is directly proportional to $\vec{H}$. That is, $\kappa_m = B/H$. But the permeability κ_m of the ratio B/H is not a constant for ferromagnetic materials. It means $\vec{B}$ is not proportional to $\vec{H}$ for ferromagnetic materials. In order to know the characteristics

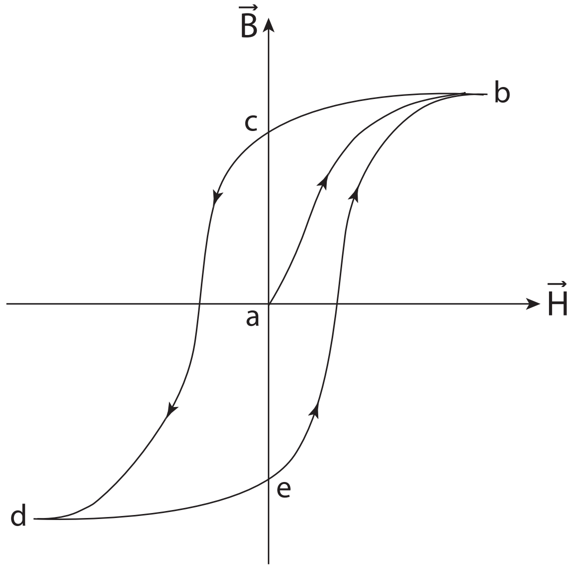


Figure 5.7: The hysteresis curve, shows the variation of magnetization of a specimen of ferromagnetic material with the variation of applied field.

between $\vec{B}$ and $\vec{H}$ for a ferromagnetic material, let us insert an iron core in the toroid like as Fig. 5.6. A curve can be plotted by putting $\vec{B}$ as ordinate and $\vec{H}$ as abscissa. In the beginning, the initial current is zero in the coil and that the iron core is demagnetized. So, initially both H and B are zero. We increase H by increasing increasing current in the solenoid. The induction field B will increase rapidly following a path *ab*, until it reaches a saturation value as it is shown in Fig. 5.7. Now we decrease the current to zero, see the net magnetized field B does not retrace its original path *ba*, but it follows the path *bc*. Notice that the iron remains magnetized (indicated value *ac*) even when the applied field H is zero. We then reverse direction of the current in the toroid, we reach a saturated magnetization in the opposite direction (point *d*). Again we reduce the current to zero, we see that the magnetization does not retrace to *da*, but it follows the path *dc* and the sample retains a permanent magnetization (indicated value *ae*). Now we reverse the current in the opposite direction until the path reaches at the saturation value (point *b*). If we supply current in the toroid to make a contour, the path is followed the trace *bcdeb*. The lack of retractibility of the net magnetization field B is called *hysteresis* and the loop *bcdeb* is called *hysteresis loop*.

At point *c* and *e* the specimen is magnetized even there is no current in the toroid. The value *ac* or *ae* is called *residual magnetization*. Furthermore, the iron 'remembers' how it became magnetized, a negative current produces a magnetization different from a positive one. This 'memory' is essential to the operation of magnetic storage of information such as on cassette tapes, computer disks, credit cards etc.

Domains:

A material such as iron is composed of a large number of microscopic crystals. The regions of roughly 0.01 mm in size in which the coupling of atomic magnetic dipoles produces essentially perfect alignment to all the atoms within each crystal. This particular region is called *magnetic domains*. There are many domains, each with its dipoles pointing in a different direction, and the net result of adding these dipole moments in an unmagnetized ferromagnetic substance gives a magnetization of zero. When the ferromagnet is placed in an external field, two effects may occur: (1) dipoles outside the walls of domains that are aligned with the field can rotate into alignment, in effect allowing such domains to grow at the expense of neighboring domains: and (2) the dipoles of nonaligned domains may swing entirely into alignment, with the applied field. In either case, there are now more dipoles aligned with the field, and material has a large magnetization. When the field is removed, the domain walls do not move completely back to their former positions, and the material retains a magnetization in the direction of the applied field.

5.6 Examples

Example 5.1 A proton is in a magnetic field of strength B = 1.5 T. The magnetic dipole moment of the proton is initially antiparallel to the direction of $\vec{B}$. How much external work must be done to reverse the direction of the magnetic dipole moment of the proton? Dipole moment of a proton is $1.41 \times 10^{-26}\ J/T$.

Solution: The energy of interaction of a magnetic dipole with a magnetic field is given by Eq. 5.6

$$U = -\vec{\mu}.\vec{B},$$

Since $\vec{\mu}$ and $\vec{B}$ are antiparallel to each other (angle between them is 180^o), the initial energy U_i is

$$U_i = -\vec{\mu}.\vec{B} = -\mu B\ \cos 180^o = +\mu B.$$

When magnetic dipole moments become parallel to B, then final energy U_f is

$$U_f = -\vec{\mu}.\vec{B} = -\mu B\ \cos 0^o = -\mu B.$$

The external work done on the system is equal to the change in energy:

$$W = U_f - U_i = -\mu B - \mu B = -2\mu B = -2(1.41\times 10^{-26}\ J/T)(1.5\ T) = -4.23\times 10-26\ J = -0.26\ \mu eV.$$

(Note: Here the environment does negative work on the system, the system does positive work on the environment. The energy might be transmitted to the environment in the form of electromagnetic radiation and would have a frequency of 64 MHz.)

Example 5.2 A 300 turns of circular coil of radius 3 cm carries a current of 85 μA. (a) What is the magnetic dipole moment of this coil? (b) The magnetic dipole moment of the coil is lined up with an external magnetic field whose strength is 0.85 T. How much work would be done by an external agent to rotate the coil through 180^o.

Solution: (a) The magnitude of the magnetic dipole moment is

$$\mu = NiA = Ni(\pi r^2) = 300(85\times 10^{-6}\ A)(\pi)(0.03\ m^2) = 7.21\times 10^{-5}\ A.m^2 = 7.21\times 10^{-5}\ J/T.$$

(b) The external work done is equal to the increase of potential energy

$W = \Delta U = -\mu B\, \cos 180^o - (-\mu B\, \cos 0^o) = 2\mu B = 2(7.21\times10^{-6}\, J/T)(0.85\, T) = 1.23\times10^{-4}\, J.$

Example 5.3 Calculate the change in magnetic moment of a circulating electron in an applied field B_0 of 2.0 T acting perpendicular to the plane of the orbit. Take $r = 5.29\times10^{-11} m$ for the radius of the orbit, corresponding to the normal state of an atom of hydrogen.

Solution: We have from Eq. 5.8

$$\mu = \frac{erv}{2} = \frac{er^2\omega}{2},$$

where $v = \omega r$. The change in magnetic moment corresponding to a change in the angular frequency ($\Delta\omega = \pm\frac{eB_o}{2m}$) is given by

$$\Delta\mu = \frac{er^2\Delta\omega}{2} = \frac{1}{2}er^2\left(\pm\frac{eB_o}{2m}\right) = \pm\frac{e^2B_or^2}{4m}$$

$$= \pm\frac{(1.6\times10^{-19}\, C)^2(2.0\, T)(5.29\times10^{-11}\, m)^2}{4(9.1\times10^{-31}\, kg)}$$

$$= \pm3.9\times10^{-29}\, J/T$$

$$= \pm\frac{3.9\times10^{-29}\, J/T}{9.27\times10^{-24}\, J/T}\mu_B$$

$$= 4.2\times10^{-6}\mu_B.$$

We see that the effect amounts to only about 4.2×10^{-6} of the magnetic moment.

Example 5.4 The magnetic susceptibility of a medium is 948×10^{-11}. Calculate the permeability (or absolute permeability) and relative permeability.

Solution: Here the susceptibility $\chi_m = 948\times10^{-11}$.

The permeability,

$$\kappa_m = \mu_o(1+\chi_m) = 4\pi\times10^{-7}(1+948\times10^{-11}) = 4\pi\times10^{-7}(949\times10^{-11})$$

$$= 1.186\times10^{-14}\ \text{h/m}.$$

The relative permeability,

$$\kappa_r = \frac{\kappa_m}{\mu_o} = \frac{4\pi\times10^{-7}(949\times10^{-11})}{4\pi\times10^{-7}}$$

$$= 949\times10^{-11}.$$

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